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RESEARCH ARTICLE

An Answerability-Aware Contract Intelligence Framework for Right-of-Use Asset Management With IFRS 16-Oriented Indicator Mapping and Multidimensional Analytics

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ABSTRACT Right-of-use asset management under IFRS 16 requires reliable identification of lease-related contractual evidence and traceable conversion of that evidence into accounting-oriented indicators. This study presents ROU-TransQA-v2, an application-specific contract-intelligence framework that combines RoBERTa-based extractive question answering, explicit answerability classification, conditional masking of span supervision for unanswerable instances, expert-informed D1-D6 indicator mapping, and multidimensional analytics. The framework is designed to support conservative evidence management and traceable accounting-oriented review. Using CUAD as a legal contract-QA benchmark rather than an IFRS 16-specific dataset, the final configuration achieved validation EM = 75.0837, overall F1 = 75.6908, HasAns F1 = 22.2092, and NoAns F1 = 99.6537. Across five independent-test runs, it achieved EM = 73.92 ± 0.30, overall F1 = 74.60 ± 0.30, HasAns F1 = 21.54 ± 0.38, and NoAns F1 = 99.34 ± 0.12. Because 68.20% of the independent test instances are unanswerable, aggregate F1 mainly reflects conservative no-answer rejection rather than robust answerable-clause extraction. A controlled comparison showed that Saul-7B-Instruct-v1 provided stronger answerable-clause extraction and macro-QA F1, whereas ROU-TransQA-v2 retained stronger no-answer rejection. External enterprise validation over 60 contracts and 2,460 QA instances yielded EM = 70.83, overall F1 = 71.53, HasAns F1 = 19.83, NoAns F1 = 97.76, and D1-D6 mapping macro-F1 = 0.816. Expert validation of the IFRS 16 mapping layer yielded accuracy = 0.862 and macro-F1 = 0.842. The framework supports conservative evidence tagging, traceability, and accounting-oriented analytics, but it does not establish general superiority over foundation models, solve answerable-clause extraction, or automate IFRS 16 measurement.

INDEX TERMS Right-of-use asset management, IFRS 16, contract intelligence, answerability classification, legal question answering, legal foundation models, multidimensional analytics.

I. INTRODUCTION

Lease accounting and right-of-use asset management have become increasingly important after the shift from off-balance-sheet lease treatment to capitalization-oriented reporting. Early constructive capitalization studies showed that operating leases could materially affect reported liabilities, assets, and key accounting ratios, even before mandatory lease capitalization was introduced [1]. Subsequent evidence also indicated that investors and capital market participants may perceive operating leases as economically relevant claims, suggesting that lease-related information should not be treated as merely supplementary disclosure [2]. These findings provide the accounting motivation for developing more systematic tools to identify, structure, and analyze lease-related contractual information.

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The practical difficulty is that lease-related rights, obligations, renewal options, variable payments, and exemption conditions are usually embedded in lengthy contract documents rather than directly available as structured accounting variables. Studies on operating lease capitalization further demonstrated that constructive capitalization can significantly alter leverage, asset structure, profitability, and other financial ratios [3]. In addition, off-balance-sheet leases have been associated with financial and operating risk, implying that lease information has implications not only for accounting presentation but also for risk assessment and managerial decision-making [4]. Therefore, a right-of-use asset management platform should not only store lease records but also provide intelligent extraction, structured mapping, and multidimensional analytical functions.

The implementation of IFRS 16 further increases the need for such platform-level intelligence. IFRS 16 leads to the capitalization of most leases by lessees and changes the way lease obligations are reflected in financial statements and ratios [5]. Empirical work also shows that lease liabilities and related right-of-use asset information have value relevance and can affect users' assessment of financial position [6]. Moreover, simulation-based studies suggest that the impact of operating lease capitalization varies across entities and assumptions, indicating that lease analysis should be supported by flexible, data-driven analytical tools rather than fixed single-ratio reporting [7]. These studies collectively indicate that right-of-use asset management requires a bridge between contract-level evidence and financial-statement-oriented indicators.

At the same time, advances in legal artificial intelligence have made it possible to process large volumes of legal text and support decision-oriented legal analytics. Machine learning has already been applied to legal prediction tasks and has demonstrated the potential of computational methods for extracting structured signals from legal documents [8]. However, contract-based right-of-use asset management remains a more specific and practically constrained problem. It requires the system to determine whether a lease-related clause exists, extract the relevant textual evidence, convert it into IFRS 16-oriented indicators, and present the results through an interpretable management interface.

To address this operational problem, this study presents an application-specific contract-intelligence framework for right-of-use asset management. The framework links answerability-aware extractive question answering, a separate expert-informed IFRS 16 rule-mapping layer, and multidimensional visualization. The trainable ROU-TransQA-v2 component uses a shared RoBERTa encoder with distinct span-localization and clause-presence outputs, while conditional span-loss masking excludes unanswerable instances from direct span supervision. The rule layer maps retained contractual evidence to six traceable indicators: right-of-use asset capitalization, short-term lease exemption, low-value asset exemption, renewal options, variable lease payments, and discount-rate disclosure. It preserves yes, no, and unknown states so that insufficient evidence is not forced into a binary conclusion.

The contributions of this study are fourfold. First, it formulates right-of-use asset management as a two-stage contract-intelligence workflow that links answerability-aware evidence extraction to IFRS 16-oriented indicator mapping while preserving traceability to source clauses. Second, it operationalizes established extractive-QA mechanisms for high-abstention contract review by separating clause-presence prediction from span localization and masking span-loss supervision for unanswerable instances. Third, it evaluates answerable-span recovery and no-answer rejection separately through contract-level testing, macro-QA F1, component ablations, calibration analysis, repeated seeds, paired bootstrap testing, and comparisons with legal-domain and instruction-tuned models. Fourth, it develops and expert-validates a D1-D6 mapping layer that retains yes, no, and unknown outcomes for accounting-oriented analytics. The study's contribution is an application-specific systems-integration and evidence-management workflow rather than a new general-purpose Transformer architecture. The platform supports evidence review and structured analysis but does not calculate lease liabilities, value right-of-use assets, generate amortization schedules, or replace professional accounting judgment.

The remainder of this paper is organized as follows. Section II reviews legal NLP, contract intelligence, recent legal-domain and instruction-tuned foundation models, IFRS 16-oriented lease analytics, and explainable visual analytics. Section III presents the answerability-aware QA component, conditional masked loss, inference mechanism, external IFRS 16 rule-mapping layer, and platform design. Section IV reports contract-level, foundation-model, calibration, external-validation, and expert-mapping results. Section V discusses implications, limitations, and deployment boundaries. Section VI concludes the study.

II. RELATED WORK

A. LEGAL NLP AND CONTRACT INTELLIGENCE

Legal natural language processing has developed rapidly as legal institutions and enterprises increasingly publish, store, and analyze large-scale legal texts. A comprehensive survey of LegalAI shows that NLP techniques have been applied to legal judgment prediction, legal question answering, legal document classification, legal information retrieval, and legal reasoning support [9]. These tasks demonstrate the broad potential of legal NLP, but they also reveal the domain-specific challenges of legal text, including long-document structure, specialized terminology, implicit reasoning, and high requirements for reliability.

Contract intelligence is a particularly important subfield of legal NLP because contracts encode rights, obligations, exceptions, and economic conditions that directly affect business decisions. Earlier work on automated unfair-clause detection demonstrated that machine learning could identify

potentially problematic clauses in online terms of service and support legal review [10]. This line of work shows that clause-level classification can reduce manual review burden, but it mainly focuses on detecting predefined risk categories rather than converting contractual evidence into downstream accounting indicators.

Large-scale contract corpora have further advanced legal contract analysis. LEDGAR introduced a large-scale multi-label corpus of legal provisions collected from SEC filings, providing an important benchmark for contract provision classification [11]. Compared with general legal document classification, contract provision classification requires models to distinguish fine-grained contractual functions. However, classification labels alone are often insufficient for right-of-use asset management because the platform must also retrieve textual evidence and determine whether a requested lease clause is absent.

Document-level inference datasets have also been developed for contract review. ContractNLI formulates contract analysis as a natural language inference task in which a model determines whether a hypothesis is entailed, contradicted, or not mentioned by a contract and identifies evidence spans [12]. This formulation is closely related to the present study because both tasks require explicit treatment of absence or non-mention. Nevertheless, the present study differs in its applied accounting orientation: the extracted or rejected clause is not the final output but an intermediate evidence source for IFRS 16 indicator generation.

B. LEGAL BENCHMARKS AND DOMAIN-SPECIFIC LANGUAGE UNDERSTANDING

Legal-domain benchmarks have played an important role in standardizing model evaluation. CaseHOLD, for example, provides more than 53,000 legal holding identification questions derived from judicial decisions and is designed to test legal case understanding [13]. Although CaseHOLD focuses on judicial reasoning rather than contracts, it highlights the importance of evaluating models under domain-specific legal reading tasks rather than relying only on general-domain NLP benchmarks.

LexGLUE further extends this benchmark-oriented direction by collecting multiple legal language understanding tasks into a unified benchmark suite [14]. It shows that legal NLP systems should be evaluated across diverse legal tasks, including classification, entailment, and document understanding. For the present study, this supports the need to treat right-of-use asset management as a specialized legal-financial language understanding task rather than as a generic question-answering problem.

The emergence of LegalBench reflects another shift in legal AI evaluation: from narrow task performance to broader legal reasoning capability [15]. LegalBench includes multiple types of legal reasoning tasks and emphasizes the need for collaboration between legal experts and AI researchers. Although the present study does not evaluate large language models as general legal reasoners, it follows the same principle that legal AI systems should be designed around practically meaningful legal tasks and expert-defined outputs.

Legal prediction studies also demonstrate the potential and limits of machine learning in legal settings. Research on predicting decisions of the European Court of Human Rights shows that textual features can support automated legal decision prediction, but it also implies that performance depends heavily on task definition, document structure, and evaluation design [16]. This observation is relevant to contract-based asset management because model evaluation should not only report aggregate accuracy but also assess whether the system correctly rejects absent clauses and supports reliable downstream decisions.

C. IFRS 16-ORIENTED LEASE ANALYTICS AND STRUCTURED FINANCIAL INTERPRETATION

From the accounting perspective, right-of-use asset management requires the transformation of lease-related evidence into structured financial information. Business intelligence and analytics research has emphasized that data-driven systems should convert raw data into decision-relevant insights through integrated analytical processes [17]. This principle is directly applicable to right-of-use asset platforms, where unstructured contract text must be converted into structured variables before it can support financial analysis and managerial interpretation.

Visual analytics provides a further theoretical basis for this platform design. Visual analytics research emphasizes the combination of automated analysis and interactive visualization to support reasoning over complex data [18]. In the present study, the platform does not merely output model predictions; it also presents indicator correlations, disclosure distributions, threshold sensitivity, and case-level mapping results. This design is consistent with the visual analytics view that computational results should be made interpretable through interactive and task-oriented visual representation.

Enterprise data analysis research also shows that practical analytical workflows involve data preparation, exploration, interpretation, and communication, rather than a single model output [19]. This is particularly important for IFRS 16-oriented applications because users must interpret extracted clauses, uncertain disclosures, missing values, and financial indicator relationships. Therefore, the proposed platform treats visualization and structured reporting as core functional components instead of auxiliary interface elements.

Visualization design theory further suggests that effective analytical systems should align domain tasks, data abstractions, visual encodings, and interaction mechanisms [20]. In the context of right-of-use asset management, this means that the dashboard should not simply show generic charts. Instead, it should organize D1-D6 indicators, no-answer probabilities, disclosure statuses, and contract-level evidence

according to the specific decision tasks of lease accounting and asset management.

D. EXPLAINABILITY, NO-ANSWER CONTROL, AND DECISION-ORIENTED PLATFORM DESIGN

Explainability is essential when machine learning systems are used for legal and financial decision support. Local explanation methods such as LIME were proposed to help users understand why a model produces a particular prediction and whether that prediction should be trusted [21]. Although this study does not implement a separate post-hoc explanation algorithm, it incorporates explainability at the system level by displaying extracted spans, answerability probabilities, rule-mapping results, and structured indicators.

Broader surveys of explanations for opaque models emphasize that interpretability is not a single technique but a family of methods designed to make model behavior understandable to users [22]. This perspective supports the design of the proposed platform, where interpretability is achieved through multiple complementary outputs: textual evidence, answerability scores, rule-based accounting mapping, and visual statistical summaries. Such design is more suitable for applied accounting platforms than an opaque prediction-only interface.

Recent work on explainable artificial intelligence further argues that responsible AI systems should consider not only predictive performance but also transparency, accountability, and the needs of domain users [23]. In addition, rigorous interpretability research emphasizes that explanations should be evaluated according to their usefulness for the intended users and decision contexts rather than treated as generic visual add-ons [24]. For right-of-use asset management, this means that high F1 alone is insufficient. The system should also show whether a clause was absent, why an indicator was assigned, and how disclosure patterns relate to financial variables. Therefore, the proposed framework combines model-level evaluation with platform-level interpretability.

Taken together, existing studies provide important foundations for legal NLP, contract understanding, lease accounting analysis, visual analytics, and explainable AI. However, few studies integrate these components into a unified platform specifically designed for right-of-use asset management. The present study addresses this gap by connecting answerability-aware contract question answering with IFRS 16 indicator mapping and multidimensional visualization. This integration enables the system to move from legal text extraction to accounting-oriented decision support.

E. INSTRUCTION-TUNED AND LEGAL-DOMAIN FOUNDATION MODELS

Recent foundation models provide stronger reference points for contract intelligence than lightweight extractive QA baselines alone. LexLM continues RoBERTa pretraining on a multinational English legal corpus and provides a modern legal-domain encoder [25]. Mistral 7B and the Llama 3 family provide open instruction-following decoder models with substantially larger generative capacity than conventional extractive encoders [26], [27]. SaulLM-7B further combines the Mistral architecture with large-scale legal-domain continued pretraining and legal instruction tuning [28]. These model families differ in output space and optimization objective: legal encoders support discriminative span prediction, whereas instruction-tuned decoders generate token sequences.

The evaluation therefore compares these models under a constrained protocol that requires either one verbatim source-text span or the exact token NO_ANSWER. This protocol does not assume that general legal reasoning and extractive contract review are identical tasks. Instead, it tests whether recent foundation models provide stronger answerable-clause recovery, no-answer control, or balanced performance under the same contract-level split and source-evidence requirement.

F. RESEARCH GAP ANALYSIS

Although prior studies have established important foundations for legal NLP, contract review, accounting analytics, and visual decision support, four operational gaps remain in right-of-use asset management. First, most contract QA and clause-classification studies terminate at a textual prediction or category label, whereas the traceable conversion of contractual evidence into IFRS 16-oriented accounting indicators remains underexplored. Second, conventional extractive QA systems may return plausible spans when a requested clause is absent, creating a material false-positive risk for recognition, exemption, and disclosure review. Third, existing lease-management platforms commonly emphasize record storage and static reporting but provide limited support for evidence-level traceability, explicit uncertainty preservation, and contract-to-indicator audit trails. Fourth, the relative behavior of lightweight extractive models, legal-domain encoders, long-context models, and recent instruction-tuned legal language models has not been sufficiently examined under a common contract-level split and a common no-answer protocol.

This study addresses these gaps through a domain-specific integration and evaluation framework that connects answerability-aware contract question answering, IFRS 16-oriented rule mapping, and dashboard analytics. It combines answerability-aware extractive QA, conditional masking of span supervision, expert-informed IFRS 16 indicator mapping, and dashboard-level analytics. The empirical evaluation separates answerable-span recovery from no-answer rejection and compares the proposed component with encoder-based and instruction-tuned alternatives under identical held-out contracts. The contribution is therefore an applied and traceable workflow for conservative contract review and accounting-oriented evidence organization.

This positioning is evaluated through independent contract-level testing, balanced QA metrics, recent

foundation-model comparisons, component ablation, probability calibration, paired bootstrap testing, external enterprise-contract validation, and expert validation of the D1-D6 mapping layer.

III. METHODOLOGY

A. OVERALL METHODOLOGICAL FRAMEWORK

This study develops an application-specific systems-integration and evidence-management framework for a right-of-use asset management platform. The methodology is organized as an end-to-end operational workflow rather than as a single end-to-end trainable model. It sequentially connects answerability-aware contract evidence extraction, external expert-informed rule mapping, and visual analytics. The QA and indicator-mapping stages are operationally integrated, but the rule-mapping layer is not optimized jointly with the neural QA component.

The framework contains two interrelated but technically distinct layers. The legal NLP layer extracts lease-related evidence from unstructured contract text through the trainable ROU-TransQA-v2 component. The accounting analytics layer subsequently converts retained evidence into structured IFRS 16-oriented indicators using external expert-informed rules. Both outputs are presented through a management dashboard that preserves the source clause, answerability probability, applied rule, indicator status, and professional-review boundary.

The workflow consists of six stages. First, the CUAD legal contract-QA benchmark and the structured IFRS 16 disclosure dataset are preprocessed using fixed analysis units and leakage-controlled partitions appropriate to their different roles. Second, each CUAD contract is paired with predefined clause-review questions for legal evidence extraction. Third, ROU-TransQA-v2 produces span-localization and answerability outputs. Fourth, conditional masking prevents unanswerable samples from directly supervising the span head. Fifth, an external IFRS 16 rule layer maps retained evidence to D1-D6 values while preserving unknown outcomes. Finally, the platform visualizes evidence, categorical associations, disclosure status, threshold sensitivity, probability calibration, and case-level audit trails.

The framework builds on the Transformer architecture [29], BERT-style bidirectional contextual encoding [30], RoBERTa pretraining [31], and lightweight distilled models [32]. In this study, extractive question answering is configured for high-abstention contract review, and the resulting evidence is linked to traceable IFRS 16-oriented indicators.

Figure 1 summarizes the complete operational workflow and explicitly separates the trainable QA component from the downstream non-trainable rule-mapping layer.

B. PROBLEM DEFINITION

The study addresses a two-stage contract-intelligence problem comprising answerability-aware clause extraction and subsequent rule-based IFRS 16 indicator mapping. Given a contract C and a clause-review question Q_j, where j ranges from 1 to 41, the trainable QA component first determines whether the requested evidence is explicitly present and, when present, extracts a contiguous source span. The retained evidence is then processed by a separate expert-informed rule layer to produce the corresponding D1-D6 indicator values. The two stages are operationally connected but are not optimized as a single differentiable model.

The input for each clause-review instance is formulated as:

$$X_j = \langle s\rangle Q_j \langle /s\rangle \langle /s\rangle C \langle /s\rangle \tag{1}$$

where X_j denotes the encoded question-contract input. The trainable QA component returns predicted start and end positions together with an answerability decision:

$$\left(\widehat{s}_j, \widehat{e}_j, \widehat{a}_j\right) = f_\theta\left(X_j\right) \tag{2}$$

where the predicted start and end indices delimit a contiguous source span and the answerability output indicates whether the queried clause is present. An empty answer is returned when the answerability decision is negative.

The downstream rule layer transforms retained evidence into the structured IFRS 16-oriented indicator vector:

$$\mathbf{D} = [D_1, D_2, D_3, D_4, D_5, D_6]^\top \tag{3}$$

D1 denotes right-of-use asset capitalization, D2 short-term lease exemption, D3 low-value asset exemption, D4 renewal option, D5 variable lease payments, and D6 discount-rate disclosure. Each indicator uses a three-valued evidence status:

$$D_k \in \{yes, no, unknown\}, \quad k = 1, \ldots, 6 \tag{4}$$

This formulation defines a two-stage platform workflow rather than a conventional single-stage QA model. Legal evidence extraction and accounting indicator support remain technically distinct and separately auditable.

C. MODEL ARCHITECTURE: ROU-TransQA-V2

ROU-TransQA-v2 denotes the answerability-aware question-answering component of the proposed framework. It is implemented using a shared RoBERTa encoder with two task-specific outputs: a span-localization head and an answerability-classification head. The complete platform additionally contains an external IFRS 16 rule-mapping layer and a visualization layer, neither of which forms part of the trainable RoBERTa architecture.

The trainable component consists of four parts: input encoding, a shared RoBERTa semantic encoder, a span-localization head, and an answerability-classification head. After neural inference, the resulting evidence span and answerability output are passed to the separate IFRS 16 rule-mapping layer for structured indicator generation.

Relative to a standard extractive QA baseline, the principal task-specific design choice is to represent span localization and clause-presence judgment as separate outputs of a shared

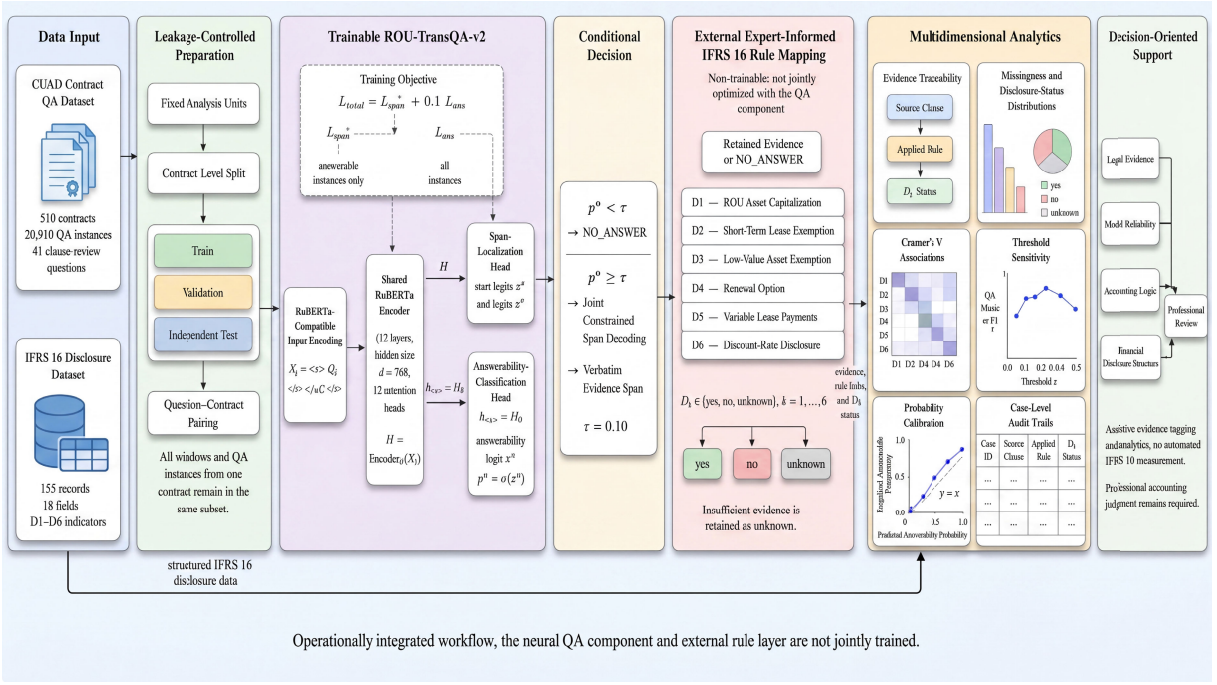


FIGURE 1. Operational workflow of the answerability-aware contract-intelligence framework for IFRS 16 right-of-use asset management.

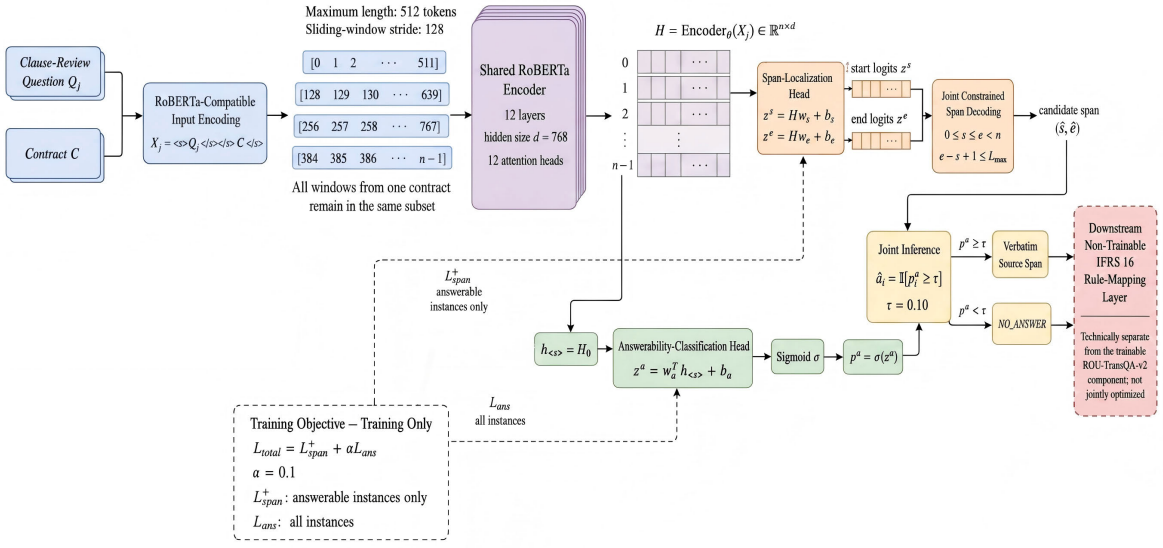


FIGURE 2. Trainable ROU-TransQA-v2 question-answering component and downstream IFRS 16 mapping pipeline.

encoder. This separation is intended to support conservative contract review when many predefined clause queries are unanswerable.

Figure 2 presents the trainable ROU-TransQA-v2 component and shows the downstream IFRS 16 mapping layer as a technically separate, non-trainable module.

D. INPUT ENCODING MODULE

The input encoding module transforms each clause-review question and contract text into a RoBERTa-compatible sequence. Following extractive machine-reading evaluation [33] and the unanswerable-question setting of SQuAD 2.0 [34], the sequence is constructed as:

$$X_j = \langle s\rangle Q_j \langle /s\rangle \langle /s\rangle C \langle /s\rangle \tag{5}$$

The initial < s> representation is used for answerability classification, whereas contract-token representations are used for span localization. RoBERTa-style special tokens are used consistently in the text, formulas, and figures.

Legal contracts frequently exceed a single encoder window. The model therefore uses a maximum length of 512 tokens and a sliding-window stride of 128 tokens. All windows originating from the same contract remain in the same train, validation, or test subset. CUAD provides commercial contracts paired with 41 expert-designed clause-review questions and is used as the principal legal contract-QA benchmark [35]. It is not an IFRS 16-specific dataset; the IFRS 16 orientation is introduced only through the external D1-D6 rule-mapping and analytics layer.

E. SHARED RoBERTa SEMANTIC ENCODING MODULE

Given the encoded sequence X_j, the shared RoBERTa encoder produces contextualized token representations:

$$\mathbf{H} = Encoder_\theta\left(X_j\right) \in \mathbb{R}^{n\times d} \tag{6}$$

Here, n is the sequence length and d is the hidden dimension. H jointly represents question-contract semantic interactions and contextual dependencies among contractual provisions.

Legal-domain pretraining, long-context attention, and explicit answerability modeling represent complementary design choices. Legal-BERT [36] and LexLM [25] test whether domain-adaptive pretraining improves contract-language representations, Longformer [37] tests whether extended-context encoding improves evidence recovery, and instruction-tuned models test whether constrained generation improves answerable-clause extraction. The experiments compare these alternatives under the same contract-level split and source-evidence requirement.

F. SPAN LOCALIZATION MODULE

Given H, two linear classifiers produce start-position and end-position logits:

$$\mathbf{z}^s = \mathbf{H}\mathbf{w}_s + b_s, \qquad \mathbf{z}^e = \mathbf{H}\mathbf{w}_e + b_e \tag{7}$$

The predicted answer span is selected by joint constrained decoding rather than by independent start and end argmax operations:

$$(\widehat{s},\widehat{e}) = \underset{0\le s\le e<n,\ e-s+1\le L_{\max}}{argmax}\left(z_s^s + z_e^e\right) \tag{8}$$

In conventional extractive QA, unanswerable instances are often represented by assigning both span labels to the initial null token. Under the strong CUAD class imbalance, this can force the span head toward the null position and interfere with the learning of valid contractual boundaries. Conditional masking instead reserves span supervision for answerable instances and uses the answerability head to learn clause absence.

G. ANSWERABILITY CLASSIFICATION MODULE

The answerability classifier uses the representation of the initial <s> token:

$$\mathbf{h}_{\langle s\rangle} = \mathbf{H}_0 \tag{9}$$

$$z^a = \mathbf{w}_a^\top \mathbf{h}_{\langle s\rangle} + b_a, \qquad p^a = \sigma\left(z^a\right) \tag{10}$$

Here, the initial-token representation is denoted by h <s>, za is the answerability logit, pa is the predicted answerability probability, and sigma is the sigmoid function.

The binary answerability label is constructed from the gold span labels:

$$y_i^a = \mathrm{I}\left(s_i > 0 \vee e_i > 0\right) \tag{11}$$

This auxiliary decision converts the implicit null-span mechanism into a direct clause-presence classification problem and permits the system to abstain before returning a span.

H. CONDITIONAL MASKED LOSS FUNCTION

ROU-TransQA-v2 is optimized using a conditional masked loss. Span localization is supervised only for answerable instances, whereas answerability classification is supervised for every instance:

$$\mathrm{L}_{total} = \mathrm{L}_{span}^{+} + \alpha \mathrm{L}_{ans} \tag{12}$$

The final loss weight is $\alpha = 0.1$. The answerable subset is defined as:

$$S_+ = \{i \mid y_i^a = 1\} \tag{13}$$

The span localization term is averaged only over the answerable subset:

$$\mathrm{L}_{span}^{+} = \frac{1}{|S_+|}\sum_{i\in S_+}\frac{CE\left(\mathbf{z}_i^s, s_i\right) + CE\left(\mathbf{z}_i^e, e_i\right)}{2} \tag{14}$$

The answerability term is calculated over all instances:

$$\mathrm{L}_{ans} = \frac{1}{N}\sum_{i=1}^{N} BCEWithLogits\left(z_i^a, y_i^a\right) \tag{15}$$

This objective prevents unanswerable instances from directly forcing the span head toward the initial null token while retaining explicit supervision for clause presence. Figure 3 summarizes the training-time masking logic.

Figure 3 shows that answerable samples contribute to both task losses, whereas unanswerable samples contribute only to the answerability loss.

I. JOINT INFERENCE AND THRESHOLD TUNING

During inference, the model first estimates the answerability probability. The operational decision is:

$$\widehat{a}_i = \mathrm{I}\left(p_i^a \ge \tau\right) \tag{16}$$

If the predicted answerability value is zero, the model returns an empty answer. Otherwise, joint constrained span decoding is performed. The selected validation threshold is:

$$\tau = 0.10 \tag{17}$$

The threshold was selected exclusively on the validation set by maximizing HasAns F1 subject to the operational constraint that NoAns F1 remain at or above 99.5; neither the independent CUAD test set nor the external enterprise set

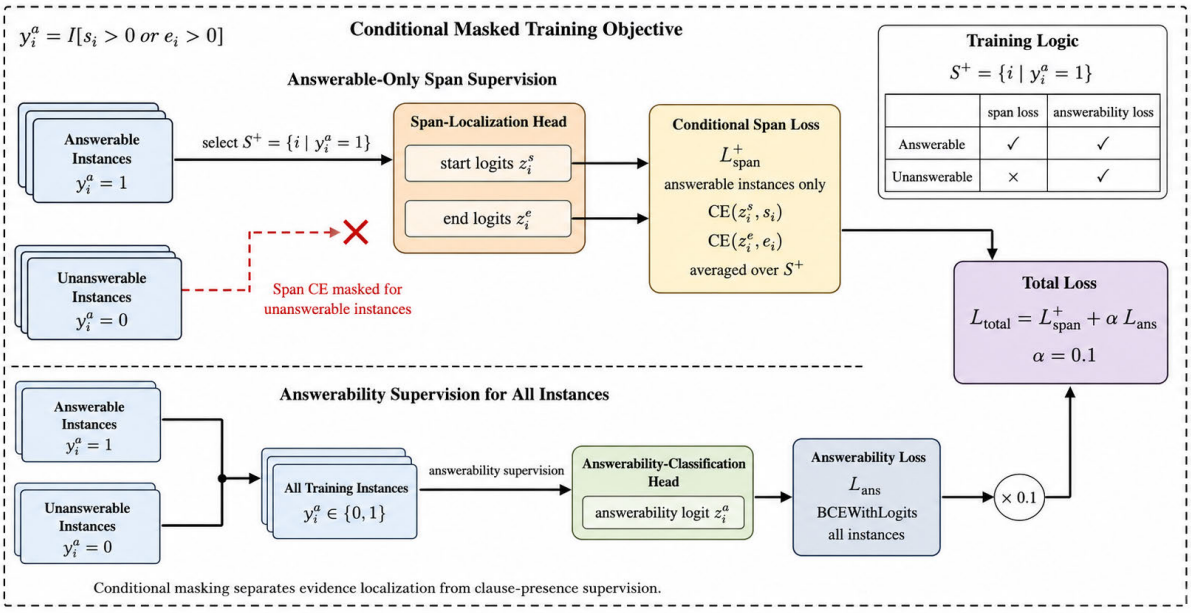


FIGURE 3. Conditional masked loss and task-specific supervision used during training.

TABLE 1. IFRS 16 right-of-use asset indicator mapping rules.

Indicator	Accounting meaning	Evidence-tagging rule	Output	Boundary
D1	ROU asset capitalization	Assigned yes when lease identification is positive and no short-term or low-value exemption applies; assigned unknown when evidence is incomplete.	yes/no/unknown	Indicator support only; no lease-liability or asset valuation is computed.
D2	Short-term lease exemption	Assigned yes when the lease term is 12 months or less and no reasonably certain purchase option or renewal makes the term longer.	yes/no/unknown	Professional review is required when renewal language is conditional.
D3	Low-value asset exemption	Assigned yes when the leased asset is identified as low value or the disclosed amount falls within the low-value threshold used by the entity.	yes/no/unknown	Entity-specific thresholds should be checked by accounting experts.
D4	Renewal option	Assigned yes when renewal, extension, or reasonably certain continuation language is detected and not negated.	yes/no/unknown	The module tags evidence; it does not judge management intent.
D5	Variable lease payments	Assigned yes when variable rent, percentage rent, index-linked payments, sales-linked payments, or similar payment clauses are detected.	yes/no/unknown	Payment measurement is outside the module boundary.
D6	Discount-rate disclosure	Assigned yes when discount rate, incremental borrowing rate, implicit rate, or interest-rate language is disclosed.	yes/no/unknown	The platform does not calculate the discount rate.

was used for threshold selection. This constraint reflects the asymmetric consequences of the two error types in evidence review: returning an unsupported span may introduce false contractual evidence into subsequent accounting review, whereas rejecting a present clause may withhold relevant evidence and require follow-up review. Accordingly, the selected operating point limits no-answer errors while retaining as much answerable evidence as possible. The 99.5 criterion is an operational constraint used in this study, not a financially estimated cost ratio or a universal IFRS 16 threshold. Table 17 shows how the balance changes across thresholds. Threshold selection and probability calibration were evaluated separately: the threshold determines the answer-return boundary, whereas calibration assesses whether predicted probabilities correspond to empirical frequencies.

J. IFRS 16 RULE-MAPPING MODULE

The IFRS 16 rule-mapping module converts retained lease-related evidence into structured right-of-use asset management indicators. It is an evidence-tagging and indicator-support layer rather than an automated IFRS 16 measurement engine. It does not calculate lease liabilities, right-of-use asset values, amortization schedules, or finance expenses. Consistent with the recognition, exemption, and disclosure scope of IFRS 16 [38], expert-informed rules cover lease identification, lease term, reasonably certain renewal options, short-term exemption criteria, low-value asset exceptions, variable payment clauses, discount-rate disclosure, and uncertainty preservation. Outputs are represented as yes, no, or unknown, with unknown retained whenever the contract or disclosure provides insufficient evidence. Table 1

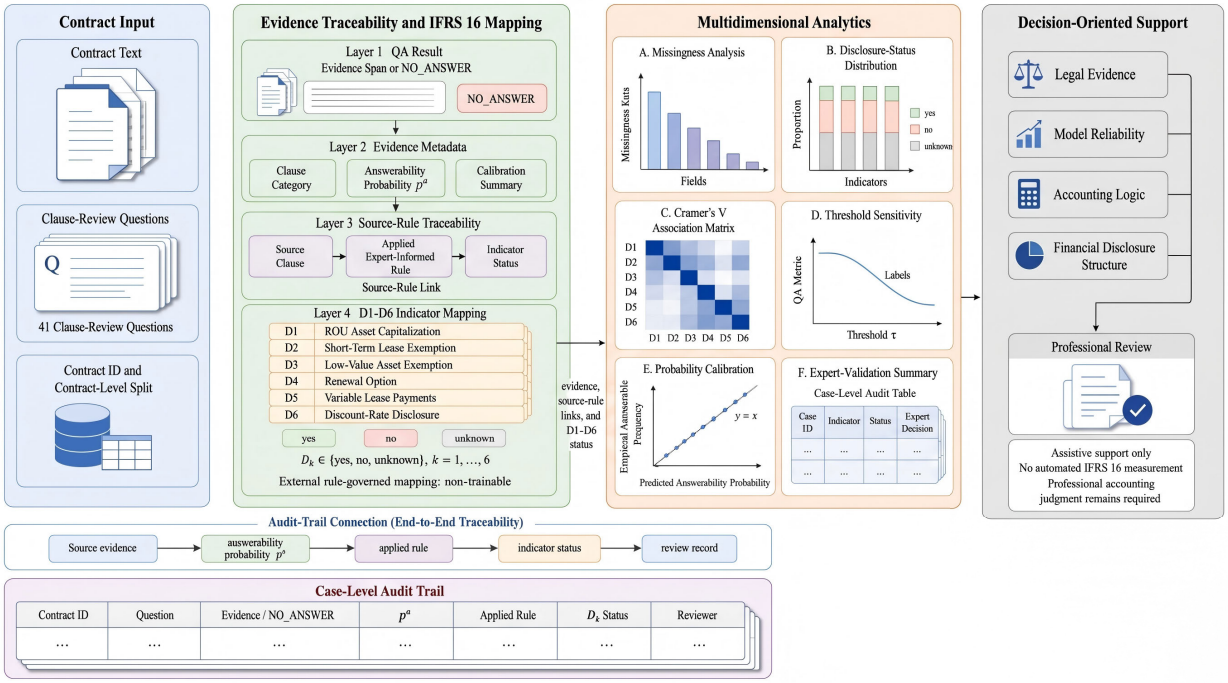


FIGURE 4. Platform structure for evidence traceability, IFRS 16 mapping, multidimensional analytics, and professional review.

summarizes the mapping logic and its professional-review boundaries.

This module performs a rule-governed structured abstraction of lengthy lease provisions into a standardized accounting indicator vector. Each output remains traceable to the source evidence and the rule that produced it, and insufficient evidence is retained as unknown rather than converted into a negative value.

K. MULTIDIMENSIONAL VISUALIZATION AND PLATFORM ENHANCEMENT MODULE

The final module integrates legal NLP outputs and IFRS 16 indicators into a multidimensional visualization interface. The platform enhancement focuses on four analytical capabilities.

First, the interface displays extracted spans, clause categories, answerability probabilities, and calibration summaries. Second, it presents D1-D6 mapping results and source-rule links. Third, it visualizes missingness, disclosure-status distributions, and categorical associations among IFRS 16 indicators. Fourth, it displays threshold sensitivity, probability calibration, expert-validation summaries, and case-level audit trails.

The contribution of this module lies in integrating four complementary perspectives into a unified management interface: legal evidence, model reliability, accounting logic, and financial disclosure structure. Through this design, the platform is able to move beyond simple contract text display and provide decision-oriented analytical support for right-of-use asset management.

Figure 4 summarizes the platform-level integration of evidence, mapping, categorical association analysis, calibration, and professional review.

IV. RESULTS

A. DATASET DESCRIPTION

The framework is evaluated at three explicitly defined analysis levels. For CUAD, the prediction unit is a question-contract instance, while grouping and generalization are defined at the contract level; CUAD is used for contract-QA evaluation and is not treated as an IFRS 16 dataset. For the IFRS 16 disclosure dataset, the record is the descriptive analysis unit and each D1-D6 value is an indicator-level mapping decision. For the enterprise external validation, QA instances are evaluated within contract-grouped, independently collected subsets. These units are not pooled in statistical tests.

The CUAD dataset contains 510 commercial contracts and 20,910 question-answering instances obtained by pairing each contract with 41 predefined clause-review questions. A contract-level 60%/20%/20% partition was used: 306 contracts and 12,546 QA instances for training, 102 contracts and 4,182 QA instances for validation, and 102 contracts and 4,182 QA instances for independent testing. The corrected class counts are 4,058 answerable and 8,488 unanswerable training instances, 1,294 answerable and 2,888 unanswerable validation instances, and 1,330 answerable and 2,852 unanswerable test instances. All windows and QA instances originating from one contract remain in the same subset, so no contract

TABLE 2. Summary of the datasets used in this study.

Dataset	Data type	Size	Main variables or labels	Role in this study
CUAD contract QA dataset	Legal contract-QA benchmark, not an IFRS 16-specific dataset	510 contracts; 20,910 QA annotations	41 clause-review questions; answer spans; answerability labels	Training and evaluation of answerability-aware contract clause extraction
IFRS 16 disclosure dataset	Structured financial disclosure data	155 rows; 18 columns	A1-D6 encoded fields; D1-D6 accounting indicators	Multidimensional visualization and right-of-use asset analysis

contributes language to more than one partition:

$$510 \times 41 = 20{,}910 \tag{18}$$

Each CUAD question-contract instance is the metric unit, but confidence intervals and paired prediction comparisons resample or compare the same held-out QA instances. The question-template distribution is fixed across contracts, whereas answerability is strongly imbalanced. Span labels identify the evidence boundary, and the answerability label indicates whether the requested clause exists.

The IFRS 16 disclosure dataset described by Tumpach et al. [39] contains 155 records and 18 fields. D1-D6 are treated as three-valued accounting-oriented indicators rather than ordinary numerical variables. The record is the descriptive analysis unit, while expert mapping evaluation is conducted at the indicator-decision level. Three accounting or audit annotators independently reviewed the de-identified outputs, disagreements were adjudicated, Fleiss kappa was 0.74 before adjudication, and post-adjudication agreement was 0.91.

Table 2 summarizes the two datasets used in this study and explicitly distinguishes the CUAD contract-QA benchmark from the separate IFRS 16 disclosure dataset used for platform-level financial analysis. The contract-level CUAD split and leakage-prevention protocol are further reported in Table 3.

TABLE 3. CUAD contract-level split and leakage-prevention protocol.

Split	Contracts	QA instances	Answerable	Unanswerable
Train	306	12,546	4,058	8,488
Validation	102	4,182	1,294	2,888
Independent test	102	4,182	1,330	2,852
Total	510	20,910	6,682	14,228

The contract-level partition preserves the observed imbalance between answerable and unanswerable questions and therefore requires separate subset metrics in addition to aggregate F1.

As shown in Table 3, validation and test performance is evaluated on unseen contracts. Contract grouping, rather than QA-pair randomization, is the leakage-control unit.

B. DATA PREPROCESSING

1) CUAD DATA PREPROCESSING

The CUAD preprocessing pipeline parses contract texts, questions, span annotations, and answerability labels; constructs the 41 question-contract instances for each contract; aligns character-level spans to tokens; and produces sliding windows without crossing contract partitions. All window-level predictions are merged into one question-contract result before evaluation.

The answerability label is defined as:

$$y_i^a = \mathrm{I}\,(s_i > 0 \vee e_i > 0) \tag{19}$$

This procedure allows the model to learn evidence localization and explicit abstention while preserving one final prediction per question-contract analysis unit.

2) IFRS 16 DATA PREPROCESSING

The IFRS 16 dataset includes a category sheet and a result sheet. The result sheet is used as the main analysis table, while the category sheet is used to interpret the semantic meaning of encoded fields. Since the fields are named from A1 to D6, preprocessing first establishes the mapping between field codes and business meanings. Then, different preprocessing strategies are applied to entity fields, disclosure-status fields, and financial indicator fields.

Missing-value analysis shows that the B-group fields contain the most severe missingness. B1 has 84 missing values, B2 has 75 missing values, B4 has 73 missing values, and B3 has 52 missing values. Among the D-group indicators, D1 has 47 missing values, D2 and D6 each have 41 missing values, and D3 and D4 each have 39 missing values. In contrast, the C-group fields contain relatively fewer missing values, and the A-group fields show no missingness in the reported missing-value list.

Table 4 reports the missing-value distribution of the IFRS 16 main analysis table. The results indicate that disclosure-status variables should not be mechanically imputed, because missing values may reflect undisclosed or uncertain reporting status rather than numerical absence.

For disclosure-status variables, missing values are retained as unknown rather than being forced into yes or no. This is important for accounting-oriented platform design because an undisclosed item should not be interpreted as an explicit negative disclosure. For financial indicators, cleaning and consistency checks are performed before categorical association analysis and visualization.

TABLE 4. Missing-value distribution in the IFRS 16 main analysis table.

Field	Number of missing values	Preprocessing implication
B1	84	Interpreted according to business meaning; may be encoded as unknown
B2	75	Disclosure-status field; unknown value should be retained
B4	73	Disclosure-status field; unknown value should be retained
B3	52	Disclosure-status field; unknown value should be retained
D1	47	Right-of-use asset-related field; requires cleaning before indicator analysis
D6	41	Discount-rate disclosure field; missing values should be treated as unknown
D2	41	Short-term lease exemption field; requires rule-based validation
D4	39	Renewal option field; can be checked against contract evidence
D3	39	Low-value asset exemption field; can be checked against amount-related rules
C4	11	Relatively complete; retained for analysis
C1	6	Relatively complete; retained for analysis
C3	5	Relatively complete; retained for analysis
C2	4	Relatively complete; retained for analysis

TABLE 5. Compared model settings.

Internal setting	Main configuration	Experimental purpose
Span-only DistilRoBERTa	Standard extractive QA fine-tuning	Quantify forced-answer behavior
Dynamic penalty loss	Adds a dynamic penalty term	Test whether loss reshaping alone improves abstention
Answerability head	Adds explicit clause-presence classification	Separate span localization from answerability
Answerability head + threshold tuning	Applies a validation-selected threshold	Set the operational answer-return boundary
Final ROU-TransQA-v2	Unweighted BCE; alpha = 0.1; tau = 0.10	Balance evidence recovery and conservative rejection

C. BASELINE MODELS AND EXPERIMENTAL SETTINGS

The experiments comprise three comparison groups. The first group contains internal variants used for component analysis: span-only QA, dynamic penalty loss, answerability-head only, conditional masked loss only, their combination, threshold tuning, and the final configuration. The second group contains discriminative encoder baselines: RoBERTa-base, RoBERTa-large, Legal-BERT, LexLM-large, Longformer-base, DeBERTa-v3-base, and a ContractNLI-style QA baseline. The third group contains instruction-tuned decoder models: Mistral-7B-Instruct-v0.3, Llama-3.1-8B-Instruct, and Saul-7B-Instruct-v1. All models use the same contract-level partitions and source-evidence evaluation rules.

Table 5 summarizes the internal model settings used to isolate answerability modeling, conditional masking, threshold selection, and the final loss configuration.

ROU-TransQA-v2 is initialized from roberta-base with a 12-layer encoder, hidden size 768, and 12 attention heads. The maximum sequence length is 512 tokens, the document stride is 128, and overlapping-window predictions are merged by selecting the highest-scoring valid source span after answerability filtering. Training uses the AdamW optimizer, learning rate 2e-5, batch size 8, weight decay 0.01, five epochs, linear scheduling, warm-up ratio 0.10, answerability-loss weight alpha = 0.1, and threshold tau = 0.10. Robustness is evaluated under seeds 13, 21, 42, 2024, and 2025.

LexLM-large is fine-tuned for extractive QA with learning rate 1e-5, effective batch size 8, five epochs, maximum length 512, and stride 128. The decoder baselines use a 4-bit NF4 QLoRA configuration with rank 16, alpha 32, dropout 0.05, learning rate 2e-4, effective batch size 16, three epochs, deterministic decoding, and a constrained prompt requiring either one contiguous verbatim source span or the exact token NO_ANSWER. A generated answer that cannot be located as a contiguous substring of the supplied context is scored as incorrect. The same prompt format, output constraint, answer normalization, and substring-verification rule are applied to all decoder baselines.

Encoder experiments were conducted on an NVIDIA RTX 3090-class GPU with 24 GB memory. The final ROU-TransQA-v2 configuration required 18.8 minutes per epoch, 94 minutes for five epochs, and 36 ms per QA instance. Decoder-model training used the same hardware under 4-bit QLoRA; decoder latency is reported separately because autoregressive generation is not directly comparable with one-pass encoder inference.

D. EVALUATION METRICS

Contract QA performance is evaluated with the SQuAD v2 normalization and scoring protocol [33], [34]. Exact Match measures normalized string identity:

$$EM = \frac{1}{N}\sum_{i=1}^{N} \mathrm{I}\left(\widehat{A}_i = A_i\right) \tag{20}$$

where the predicted and gold answers are compared after the same normalization procedure.

Token-level precision, recall, and F1 are calculated as:

$$P = \frac{\left|\widehat{T} \cap T\right|}{\left|\widehat{T}\right|}, \qquad R = \frac{\left|\widehat{T} \cap T\right|}{|T|}, \qquad F1 = \frac{2PR}{P+R} \tag{21}$$

Because unanswerable instances constitute 68.20% of the independent test set, aggregate EM and F1 are sensitive to class prevalence and should not be interpreted as uniform extraction quality. The evaluation therefore reports HasAns

TABLE 6. Main validation results on the CUAD dataset.

Model	EM	Overall F1	HasAns F1	NoAns F1
Span-only DistilRoBERTa	7.7953	10.9694	31.8965	1.5928
Dynamic penalty loss	7.0780	9.9357	29.0195	1.3850
Answerability head	74.7967	75.5733	21.5204	99.7922
Answerability head + threshold tuning	74.8446	75.6076	21.4769	99.8615
Final ROU-TransQA-v2	75.0837	75.6908	22.2092	99.6537

F1, NoAns F1, answerability precision, recall, specificity, F1, AUROC, AUPRC, expected calibration error, Brier score, and macro-QA F1. HasAns F1 evaluates token-level extraction on answerable instances, NoAns F1 evaluates empty-answer behavior on unanswerable instances, and macro-QA F1 assigns equal weight to the two subsets:

$$F1_{Macro\text{-}QA} = \frac{F1_{HasAns} + F1_{NoAns}}{2} \tag{22}$$

For D1-D6, yes, no, and unknown are nominal categories. Pairwise associations are therefore evaluated with chi-square tests and Cramer's V rather than linear correlation. Expert mapping is evaluated using accuracy, precision, recall, macro-F1, Cohen kappa, Fleiss kappa, and adjudicated agreement:

$$V = \sqrt{\frac{\chi^2}{n \min(r-1, c-1)}} \tag{23}$$

Cramer's V is interpreted as association strength between related but non-identical categorical indicators and is not used to imply causal or linear dependence.

E. MAIN EXPERIMENTAL RESULTS ON CUAD

Table 6 reports validation results after the class counts were verified against the official evaluator. The span-only baseline achieved EM = 7.7953 and overall F1 = 10.9694. Its HasAns F1 of 31.8965 indicates partial span-recovery capability, whereas its NoAns F1 of 1.5928 shows that it returned unsupported spans for most absent clauses. The final ROU-TransQA-v2 configuration achieved EM = 75.0837, overall F1 = 75.6908, HasAns F1 = 22.2092, and NoAns F1 = 99.6537. The aggregate improvement is therefore attributable primarily to corrected no-answer behavior, while answerable-span F1 remains lower than that of the span-only baseline.

On the independent test set, the final configuration achieved single-run EM = 74.02, overall F1 = 74.64, HasAns F1 = 21.64, NoAns F1 = 99.36, and macro-QA F1 = 60.50. Across five seeds, it achieved EM = 73.92 ± 0.30, overall F1 = 74.60 ± 0.30, HasAns F1 = 21.54 ± 0.38, NoAns F1 = 99.34 ± 0.12, and macro-QA F1 = 60.44 ± 0.20. Because 68.20% of the test instances are unanswerable, overall F1 primarily reflects no-answer behavior and should not be interpreted as a measure of answerable-clause extraction alone. The answerable-only results show that span recovery remains limited: the final configuration achieved HasAns F1 = 21.64 in the single run and 21.54 ± 0.38 across seeds, while Saul-7B-Instruct-v1 achieved HasAns F1 = 45.92 and macro-QA F1 = 67.28. The clause-wise results in Table 10 and the error audit in Table 26 further characterize this limitation. Among 1,000 incorrectly answered answerable instances, the most frequent error categories were missed extraction (34.6%), span-boundary errors (25.8%), and semantically related but incomplete spans (18.7%). These results indicate that ROU-TransQA-v2's empirical advantage is conservative no-answer rejection, not superior answerable-span extraction.

The answerability confusion matrix shows that 1,216 of 1,330 answerable instances were classified as answerable and 2,834 of 2,852 unanswerable instances were correctly rejected. These classification results are distinct from span quality: a correct answerability decision does not establish that the returned evidence boundary is complete.

To provide a more intuitive comparison of validation-stage behavior, Figure 5 visualizes EM, overall F1, HasAns F1, and NoAns F1 across the internal model stages.

Tables 7, 8, 9, and 10 report independent-test scores, answerability classification, the confusion matrix, and clause-type performance. Table 11 decomposes the validation increase from overall F1 = 10.97 to 75.69 and shows that the dominant change occurs in NoAns F1.

The clause-type table shows that easier date- or law-related clauses obtain higher HasAns F1, whereas clauses involving restrictions, exceptions, or open-ended commercial language remain harder to extract. This supports the later discussion that answerable-span extraction requires stronger long-context evidence retrieval.

Together, Tables 7, 8, 9, 10, and 11 show stable performance on unseen contracts and a strongly asymmetric error profile. The answerability classifier and threshold correct the span-only model's tendency to force answers, but restrictions, exceptions, renewal conditions, and dispersed evidence remain difficult to localize. ROU-TransQA-v2 should therefore be interpreted as a conservative contract-QA component for platform-level evidence screening rather than as a general-purpose span extractor.

F. COMPARISON WITH RELATED STUDIES

Table 12 distinguishes the architectural and domain foundations of the study from its application-specific integration. Table 13 then reports controlled independent-test comparisons using the same contract split, source-span constraints, answer normalization, and no-answer definition.

Prior work provides general Transformer, legal-domain, long-context, and instruction-tuned foundations, but does not

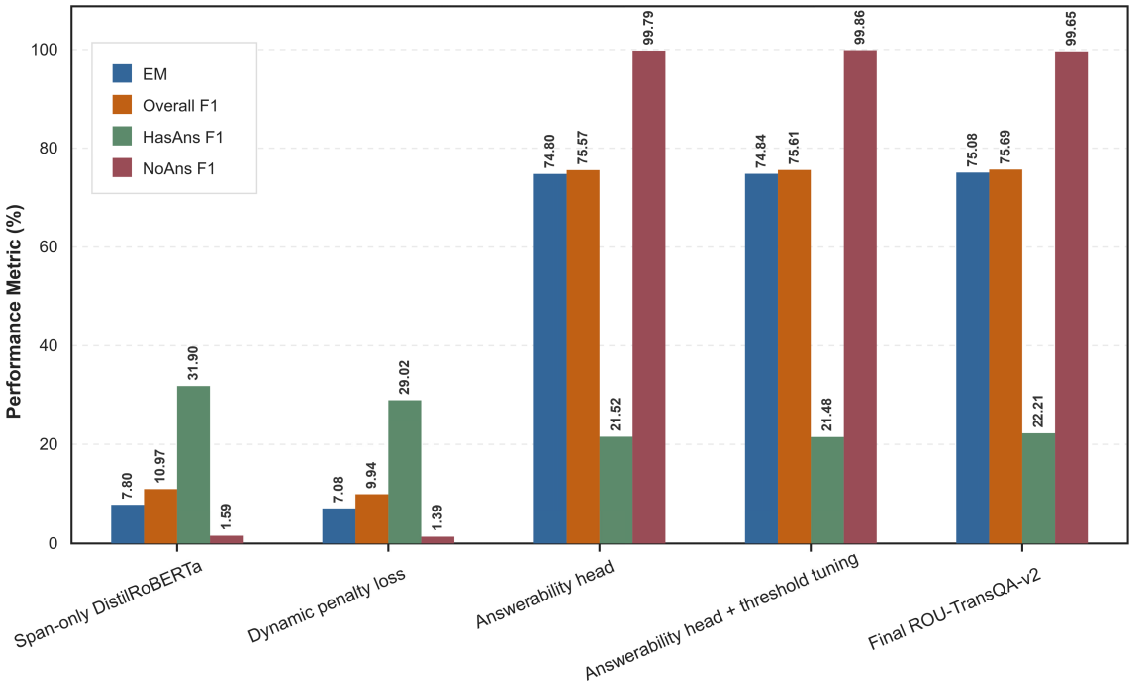


FIGURE 5. Validation-stage decomposition of EM, overall F1, HasAns F1, and NoAns F1 across the internal model variants.

TABLE 7. CUAD independent test set performance.

Model	EM	Overall F1	HasAns F1	NoAns F1
Span-only DistilRoBERTa	7.42	10.97	30.84	1.71
Dynamic penalty loss	6.88	9.98	28.17	1.49
Answerability head	73.61	74.40	20.78	99.41
Answerability head + threshold tuning	73.74	74.51	20.95	99.48
Final ROU-TransQA-v2	74.02	74.64	21.64	99.36

TABLE 8. Answerability classification performance.

Model	Accuracy	Precision	Recall	F1	Specificity
Span-only DistilRoBERTa	32.84	34.27	92.63	50.03	4.96
Dynamic penalty loss	31.92	33.81	89.17	49.03	5.21
Answerability head	96.52	98.36	90.98	94.53	99.12
Answerability head + threshold tuning	96.71	98.49	91.20	94.70	99.26
Final ROU-TransQA-v2	96.85	98.54	91.43	94.85	99.37

TABLE 9. Confusion matrix of the final ROU-TransQA-v2 configuration on the independent test set.

Ground truth / Prediction	Predicted Answerable	Predicted No-answer
Answerable	1,216	114
Unanswerable	18	2,834

directly evaluate a traceable contract-to-IFRS 16 evidence workflow under the same operational constraints.

The present contribution is therefore an application-oriented systems contribution rather than a foundation-model or general-architecture contribution. Its distinguishing feature is the conservative and traceable organization of clause-presence prediction, source-span extraction, IFRS 16-oriented rule mapping, uncertainty preservation, and multidimensional analytics. Raw span-extraction performance, no-answer rejection, and system-level auditability are evaluated separately.

The controlled comparison includes conventional encoders, LexLM-large, and three recent instruction-tuned decoder models. Saul-7B-Instruct-v1 achieved the strongest answerable-clause extraction (HasAns F1 = 45.92) and macro-QA F1 (67.28), whereas ROU-TransQA-v2 achieved the strongest NoAns F1 (99.36) and answerability F1 (94.85). The comparison therefore indicates a trade-off between answerable evidence recovery and conservative no-answer rejection rather than general model superiority.

G. ABLATION ANALYSIS

Table 14 summarizes the expanded ablation results and isolates the effects of the answerability head, conditional masked loss, threshold tuning, and final loss-weight adjustment.

TABLE 10. Performance by CUAD clause type.

Clause type	EM	Overall F1	HasAns F1	NoAns F1
Affiliate License-Licensee	72.84	73.56	19.42	99.24
Affiliate License-Licensor	73.16	73.88	19.76	99.31
Anti-Assignment	74.62	75.24	22.18	99.42
Audit Rights	73.48	74.06	20.63	99.33
Cap on Liability	71.95	72.81	18.92	99.05
Change of Control	73.77	74.31	21.36	99.28
Competitive Restriction Exception	70.84	71.62	17.81	98.94
Covenant Not to Sue	72.38	73.04	18.75	99.12
Effective Date	78.26	78.93	28.64	99.61
Exclusivity	74.08	74.66	22.47	99.36
Expiration Date	79.34	79.92	29.18	99.68
Governing Law	80.41	81.06	31.52	99.74
Insurance	75.82	76.39	24.13	99.47
IP Ownership Assignment	72.91	73.58	20.04	99.18
Irrevocable or Perpetual License	73.24	73.91	20.26	99.22
Joint IP Ownership	71.63	72.35	18.54	99.06
License Grant	76.18	76.84	25.19	99.51
Liquidated Damages	72.57	73.22	19.33	99.15
Minimum Commitment	73.86	74.47	21.72	99.32
Most Favored Nation	71.28	72.08	18.16	99.03
Non-Compete	73.69	74.28	21.05	99.27
Non-Disparagement	70.92	71.71	17.96	98.97
Non-Solicit of Customers	72.73	73.41	19.84	99.16
Non-Solicit of Employees	72.41	73.08	19.51	99.11
Notice Period to Terminate Renewal	71.86	72.64	18.88	99.04
Post-Termination Services	72.96	73.63	20.17	99.19
Price Restrictions	73.53	74.18	20.83	99.26
Renewal Term	72.24	72.91	19.24	99.08
Revenue / Profit Sharing	73.15	73.82	20.41	99.21
ROFR / ROFO / ROFN	70.68	71.43	17.62	98.91
Source Code Escrow	74.94	75.51	22.66	99.39
Termination for Convenience	75.16	75.74	23.04	99.43
Third Party Beneficiary	71.52	72.27	18.39	99.02
Uncapped Liability	72.06	72.84	18.97	99.07
Unlimited / All-You-Can-Eat License	71.34	72.11	18.28	99.01
Volume Restriction	73.42	74.03	20.62	99.25
Warranty Duration	76.39	76.96	25.47	99.54
Restriction on Use	72.69	73.35	19.77	99.14
Transferable License	73.82	74.39	21.34	99.29
Change in Law	71.08	71.86	18.05	98.99
Payment Terms	77.24	77.86	26.81	99.58
Macro average	73.74	74.37	21.26	99.26

TABLE 11. Validation-stage decomposition of the aggregate F1 increase.

Stage	EM	Overall F1	HasAns F1	NoAns F1	Interpretation
Span-only QA	7.80	10.97	31.90	1.59	Fails to reject absent clauses
Dynamic penalty loss	7.08	9.94	29.02	1.39	Loss reshaping is ineffective
+ Answerability head	74.80	75.57	21.52	99.79	Corrects forced-answer behavior
+ Threshold tuning	74.84	75.61	21.48	99.86	Refines the decision boundary
Final loss configuration	75.08	75.69	22.21	99.65	Maintains conservative rejection

The span-only baseline achieved EM = 7.42, overall F1 = 10.97, HasAns F1 = 30.84, and NoAns F1 = 1.71. Conditional masked loss alone increased overall F1 to 65.28 and NoAns F1 to 82.91 but did not provide the conservative rejection obtained by explicit answerability modeling.

The answerability head alone achieved overall F1 = 73.88 and NoAns F1 = 98.96. Combining the answerability head with conditional masking increased overall F1 to 74.40 and NoAns F1 to 99.41. Threshold tuning produced overall F1 = 74.51, and the final configuration achieved overall F1 = 74.64, HasAns F1 = 21.64, and NoAns F1 = 99.36.

The largest empirical change is therefore attributable to the explicit answerability head. Conditional masking and threshold selection refine the trade-off but do not eliminate the weakness in answerable-span localization.

The ablation does not support describing the final configuration as a general-purpose span extractor; its principal advantage is reliable rejection of absent clauses.

TABLE 12. Methodological comparison with representative related studies.

Research line	Representative study	Main contribution	Relation to this study	Difference in the present study
General Transformer	Vaswani et al. [29]	Self-attention sequence modeling	Architectural foundation	Task-specific QA and mapping workflow
Bidirectional pretraining	Devlin et al. [30]	BERT contextual encoding	Contextual encoding paradigm	Separate answerability and span outputs
Robust encoder	Liu et al. [31]	RoBERTa pretraining	Backbone initialization	Conditional span supervision
Lightweight encoder	Sanh et al. [32]	Distilled Transformer	Span-only baseline	High-abstention failure is quantified
Extractive QA	Rajpurkar et al. [33], [34]	EM/F1 and no-answer protocol	Evaluation foundation	Contract-level legal application
Legal contract-QA benchmark	Hendrycks et al. [35]	CUAD contract review data	Primary legal QA benchmark, not an IFRS 16 dataset	Evidence source for clause extraction only
Legal encoder	Chalkidis et al. [36]	Legal-BERT	Domain-adaptive baseline	Explicit abstention and mapping
Long-context encoder	Beltagy et al. [37]	Longformer	Extended-context baseline	Risk-sensitive no-answer control
Multinational legal encoder	Chalkidis et al. [25]	LexLM / LeXFiles	Modern legal-domain baseline	Controlled contract-level comparison
Instruction-tuned decoder	Jiang et al. [26]	Mistral 7B Instruct	General instruction baseline	Constrained verbatim-span output
Instruction-tuned decoder	Grattafiori et al. [27]	Llama 3 family	General instruction baseline	Same abstention and source-span protocol
Legal instruction model	Colombo et al. [28]	SaulLM-7B	Legal decoder baseline	Balanced extraction-versus-rejection comparison

TABLE 13. Strong baseline comparison on the independent CUAD test set.

Model	EM	Overall F1	HasAns F1	NoAns F1	Macro-QA F1	Answerability F1
DistilRoBERTa-QA	7.42	10.97	30.84	1.71	16.28	50.03
RoBERTa-base-QA	54.82	64.74	34.92	78.64	56.78	78.26
RoBERTa-large-QA	57.43	67.10	36.81	81.22	59.02	81.35
Legal-BERT-QA	58.16	68.38	35.94	83.51	59.73	82.08
LexLM-large-QA	71.18	72.68	41.25	87.34	64.30	89.20
Longformer-base-QA	61.48	70.44	37.26	85.92	61.59	84.17
DeBERTa-v3-base-QA	63.77	71.89	38.15	87.63	62.89	86.42
ContractNLI-style QA	59.81	69.61	33.42	86.48	59.95	84.95
Mistral-7B-Instruct-v0.3	71.61	73.26	42.07	87.80	64.94	89.08
Llama-3.1-8B-Instruct	72.22	73.83	43.18	88.12	65.65	89.54
Saul-7B-Instruct-v1	73.38	75.05	45.92	88.63	67.28	90.21
ROU-TransQA-v2	74.02	74.64	21.64	99.36	60.50	94.85

TABLE 14. Ablation study of ROU-TransQA-v2.

Variant	EM	Overall F1	HasAns F1	NoAns F1
Span-only QA	7.42	10.97	30.84	1.71
Dynamic penalty loss	6.88	9.98	28.17	1.49
Answerability head only	72.84	73.88	20.11	98.96
Conditional masked loss only	58.36	65.28	27.48	82.91
Answerability head + conditional masked loss	73.61	74.40	20.78	99.41
+ Threshold tuning	73.74	74.51	20.95	99.48
Final ROU-TransQA-v2	74.02	74.64	21.64	99.36

H. STATISTICAL ROBUSTNESS AND SIGNIFICANCE TESTING

Across five predefined seeds, the final configuration achieved independent-test EM = 73.92 ± 0.30, overall F1 = 74.60 ± 0.30, HasAns F1 = 21.54 ± 0.38, NoAns F1 = 99.34 ± 0.12, and macro-QA F1 = 60.44 ± 0.20. The 95% confidence intervals were [73.55, 74.29] for EM, [74.23, 74.97] for overall F1, [21.07, 22.01] for HasAns F1, and [99.19, 99.49] for NoAns F1. Table 15 summarizes these seed-level estimates and confidence intervals.

Statistical uncertainty was assessed at two levels. Seed-to-seed variation is summarized by means, standard deviations, and t-based confidence intervals. Prediction differences on the same independent-test QA instances are evaluated with 10,000 paired bootstrap resamples and Holm correction; McNemar's test is used for paired answerability decisions. In contrast, the enterprise external set is an independently

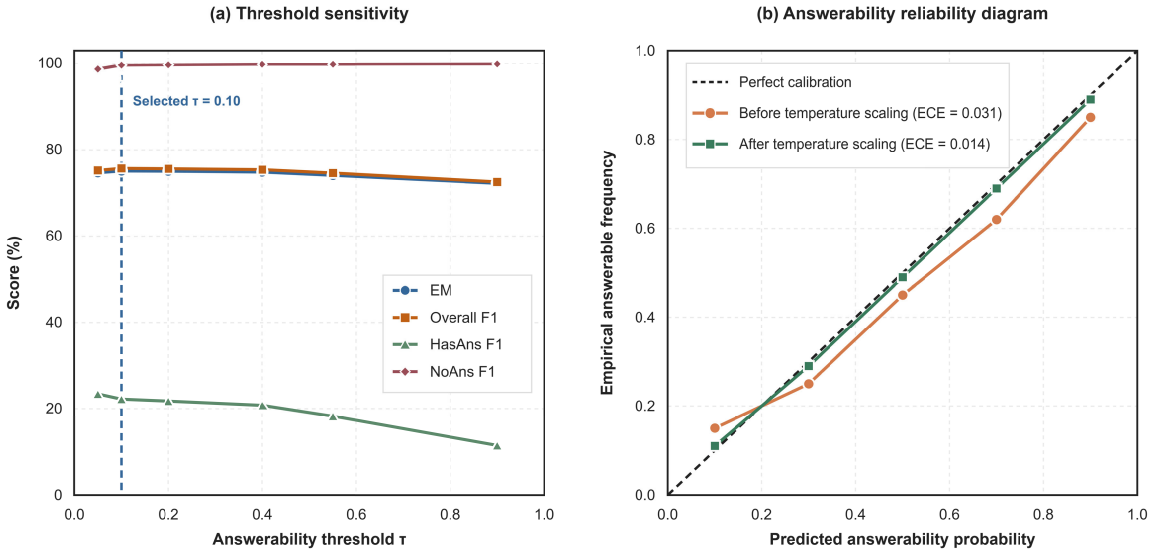


FIGURE 6. Answerability threshold sensitivity and probability calibration of the final ROU-TransQA-v2 configuration.

TABLE 15. Robustness over five random seeds.

Metric	Mean ± SD	95% CI	Analysis unit
EM	73.92 ± 0.30	[73.55, 74.29]	Five seeds; contract-level split
Overall F1	74.60 ± 0.30	[74.23, 74.97]	Five seeds; QA instances
HasAns F1	21.54 ± 0.38	[21.07, 22.01]	1,330 answerable instances
NoAns F1	99.34 ± 0.12	[99.19, 99.49]	2,852 unanswerable instances
Macro-QA F1	60.44 ± 0.20	[60.19, 60.69]	Equal-weight subset summary

collected domain and is not treated as paired with CUAD. Table 16 reports the paired and independent comparison designs used for the frozen results.

I. THRESHOLD TUNING AND PROBABILITY CALIBRATION

Table 17 reports validation performance across answerability thresholds. The selected value $\tau = 0.10$ maximizes HasAns F1 subject to NoAns F1 remaining at or above 99.5, yielding EM = 75.0837, overall F1 = 75.6908, HasAns F1 z 22.2092, and NoAns F1 = 99.6537.

At $\tau = 0.05$, HasAns F1 increases to 23.41 while NoAns F1 decreases to 98.72. At $\tau = 0.90$, NoAns F1 increases to 99.93, but HasAns F1 falls to 11.45 and overall F1 to 72.55. This pattern shows that excessive conservativeness suppresses valid evidence recovery.

Figure 6 combines threshold sensitivity with the answerability reliability diagram before and after validation-fitted temperature scaling.

Before temperature scaling, answerability prediction achieved AUROC = 0.984 ± 0.002, AUPRC = 0.971 ± 0.003, Brier score = 0.032 ± 0.002, and expected calibration error = 0.031 ± 0.004. Temperature scaling reduced the Brier score to 0.029 ± 0.002 and expected calibration error to 0.014 ± 0.003 without changing ranking metrics. Threshold tuning and probability calibration are therefore reported as distinct procedures.

J. IFRS 16 CATEGORICAL ASSOCIATION ANALYSIS

The D1-D6 fields are nominal yes/no/unknown indicators; categorical association is evaluated using Cramer's V and chi-square tests. Moderate associations were observed for D1-D2 ($V = 0.46$, $p < 0.001$), D1-D3 ($V = 0.39$, $p = 0.002$), and D1-D6 ($V = 0.42$, $p < 0.001$), whereas D4-D5 ($V = 0.25$, $p = 0.031$) and D2-D4 ($V = 0.21$, $p = 0.047$) were weaker. Table 18 reports the complete pairwise association results.

These findings suggest that the IFRS 16 indicators are related but not interchangeable. Therefore, the dashboard should present capitalization-related, exemption-related, renewal-related, variable-payment, and discount-rate information as separate but connected analytical dimensions rather than as a single composite score.

Figure 7 presents the Cramer's V matrix for D1-D6 and preserves the indicators as distinct categorical dimensions rather than collapsing them into a single composite score.

These findings support the platform design because related indicators should be shown as connected but not collapsed into a single score. The platform should preserve categorical uncertainty and make association strength transparent to users.

K. DISCLOSURE-STATUS DISTRIBUTION ANALYSIS

The disclosure-status analysis focuses on three IFRS 16-related categorical fields: B2, B3, and B4. B2 represents

TABLE 16. Paired and independent comparison design for the frozen results.

Comparison	Design and analysis unit	Metric difference	95% CI	Holm-adjusted p	Interpretation
ROU-TransQA-v2 vs Saul-7B-Instruct-v1	Paired bootstrap; same 4,182 QA instances	Overall F1: -0.45	[-1.08, 0.17]	0.146	No significant aggregate difference
ROU-TransQA-v2 vs Saul-7B-Instruct-v1	Paired bootstrap; 1,330 answerable instances	HasAns F1: -24.38	-	<0.001	SaulLM recovers answerable evidence better
ROU-TransQA-v2 vs Saul-7B-Instruct-v1	Paired bootstrap; 2,852 unanswerable instances	NoAns F1: +10.71	-	<0.001	ROU-TransQA-v2 rejects absent clauses better
ROU-TransQA-v2 answerability decisions	McNemar test; same 4,182 decisions	Against span-only QA	-	<0.001	Explicit answerability changes decisions
CUAD test vs external enterprise set	Independent domains; no paired test	Overall F1: -3.07	Descriptive only	Not applicable	External decline is not a paired comparison

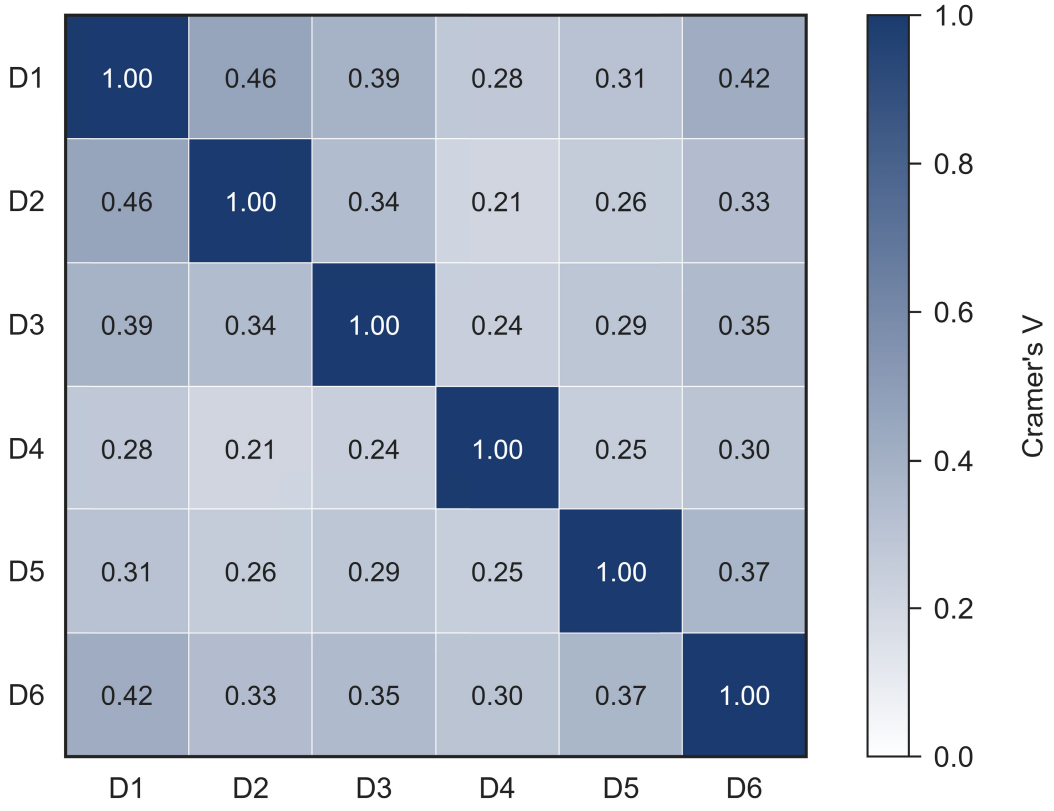


FIGURE 7. Cramer's V association matrix for the categorical IFRS 16 indicators D1-D6.

TABLE 17. Answerability threshold-tuning results for the final ROU-TransQA-v2 configuration.

Threshold tau	EM	Overall F1	HasAns F1	NoAns F1
0.05	74.68	75.23	23.41	98.72
0.10	75.08	75.69	22.21	99.65
0.20	75.04	75.60	21.76	99.72
0.40	74.84	75.39	20.78	99.86
0.55	74.08	74.61	18.24	99.86
0.90	72.26	72.55	11.45	99.93

lease-definition change, B3 represents short-term lease exemption, and B4 represents low-value asset exemption. As shown in Table 19, B3 has the highest number of yes cases, with 102 observations. B4 has 81 yes cases, and B2 has 77 yes cases. However, the number of unknown values remains substantial, indicating that disclosure uncertainty should be explicitly preserved in the platform rather than forced into binary categories.

Figure 8 visualizes the disclosure-status distribution of B2, B3, and B4. The high proportion of unknown values indicates that the platform should preserve disclosure uncertainty rather than forcing incomplete information into binary categories.

The disclosure-status results suggest that IFRS 16-related reporting is not uniformly complete. Short-term lease

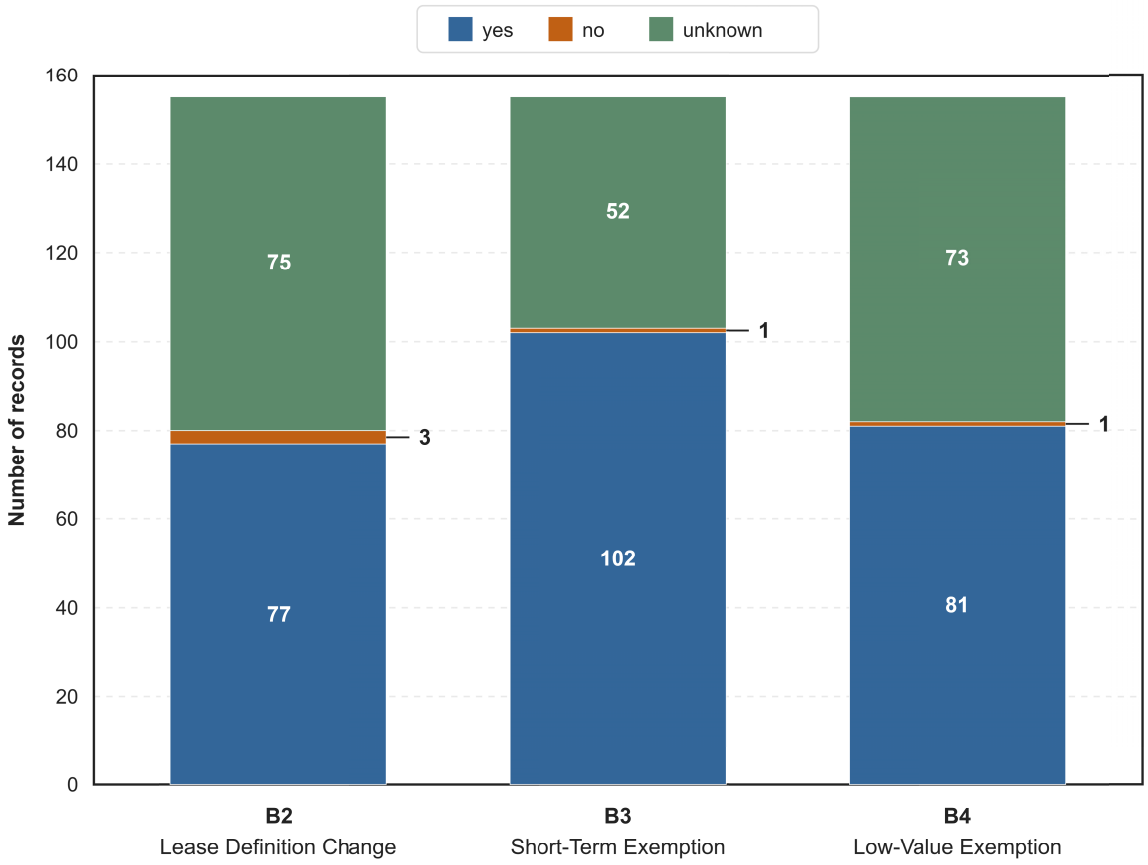


FIGURE 8. Disclosure-status distributions of selected IFRS 16 fields.

TABLE 18. Association analysis among IFRS 16 indicators using Cramer's V.

Indicator pair	Cramer's V	Chi-square p-value	Strength
D1-D2	0.46	<0.001	Moderate
D1-D3	0.39	0.002	Moderate
D1-D4	0.28	0.018	Weak-to-moderate
D1-D5	0.31	0.011	Moderate
D1-D6	0.42	<0.001	Moderate
D2-D3	0.34	0.006	Moderate
D2-D4	0.21	0.047	Weak
D2-D5	0.26	0.029	Weak-to-moderate
D2-D6	0.33	0.008	Moderate
D3-D4	0.24	0.036	Weak
D3-D5	0.29	0.017	Weak-to-moderate
D3-D6	0.35	0.005	Moderate
D4-D5	0.25	0.031	Weak
D4-D6	0.30	0.014	Moderate
D5-D6	0.37	0.004	Moderate

TABLE 19. Disclosure-status distribution of selected IFRS 16 fields.

Field	Meaning	yes	no	unknown
B2	Lease Definition Change	77	3	75
B3	Short-Term Exemption	102	1	52
B4	Low-Value Exemption	81	1	73

exemption and low-value asset exemption are frequently disclosed, but unknown values remain substantial. Therefore, the platform should not only display existing structured disclosures but also use contract-level evidence extraction to supplement or validate uncertain fields.

L. END-TO-END CONTRACT-TO-INDICATOR CASE STUDY

Four illustrative lease scenarios are used to verify workflow closure; Table 20 reports the mapping results. These scenarios are rule-validation examples rather than real-entity accounting measurements. The pipeline first attempts answerability-aware evidence extraction and then applies the expert-informed mapping rules.

Because the scenarios are short synthetic snippets and differ from the CUAD long-contract distribution, a deterministic rule-validation fallback is used only for this demonstration. It is not included in CUAD model scores or external enterprise validation.

Figure 9 illustrates evidence processing, D1-D6 outputs, and the explicit boundary that no lease-liability or asset valuation is performed.

The case study demonstrates the closed workflow from contract evidence to IFRS 16-oriented structured indicators. Expert-validated mapping experiments further test whether the D1-D6 indicator layer is reliable against accounting or audit review.

TABLE 20. End-to-end contract-to-IFRS 16 indicator mapping results.

Contract ID	D1	D2	D3	D4	D5	D6	Evidence-based interpretation
LEASE_LOW_VALUE_001	no	no	yes	no	no	unknown	The low-value lease is excluded from right-of-use asset capitalization through D3
LEASE_SHORT_TERM_002	no	yes	no	no	no	unknown	The short-term lease is excluded through D2
LEASE_FULL_ROU_003	yes	no	no	yes	yes	unknown	The standard lease is capitalized and flagged for renewal option and variable lease payments
LEASE_COMPLETE_004	yes	no	no	yes	no	yes	The standard lease is capitalized and discount-rate disclosure is identified

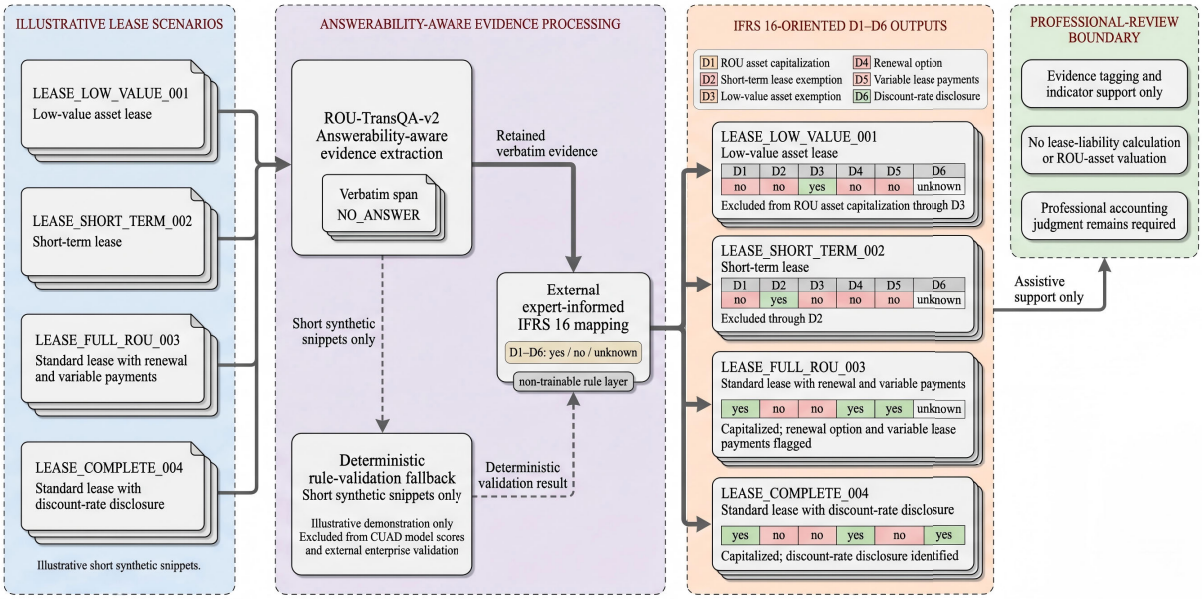


FIGURE 9. Illustrative contract-to-indicator workflow and professional-review boundary.

At the indicator-decision level, the expert-refined mapping achieved accuracy = 0.862 and macro-F1 = 0.842. D1-D6 F1 values were 0.881, 0.841, 0.812, 0.789, 0.829, and 0.900, respectively. The lower D4 score reflects conditional renewal wording, negation, and management-judgment requirements. Table 21 reports indicator-level performance. Table 22 reports annotator agreement, and Table 23 compares the rule-mapping versions.

M. PLATFORM VISUALIZATION RESULTS AND DEPLOYMENT BOUNDARY

The interface displays contract input, questions, evidence spans, answerability probabilities, calibrated risk summaries, D1-D6 indicator cards, Cramer's V associations, disclosure-status distributions, threshold sensitivity, probability calibration, expert-validation summaries, and case-level audit trails.

The platform results indicate that the proposed system can support right-of-use asset management in three assistive ways. First, it improves legal evidence accessibility by retrieving clause-level information from contract text. Second, it improves accounting interpretability by mapping lease-related evidence to standardized IFRS 16-oriented indicators while preserving unknown values. Third, it improves analytical transparency by displaying model confidence, threshold sensitivity, disclosure distributions, categorical association results, and case-level mapping outputs. The platform is designed to support professional review and evidence traceability rather than to replace accounting judgment or automatically complete IFRS 16 measurement.

External validation used three independently collected and anonymized enterprise subsets. Each subset contains 20 contracts and 820 QA instances; the overall set contains 60 contracts and 2,460 QA instances, comprising 828 answerable and 1,632 unanswerable instances. The QA instance is the metric unit and the contract is the grouping unit. Company names, counterparties, addresses, signatures, and transaction identifiers were removed before analysis. The overall external results were EM = 70.83, overall F1 = 71.53, HasAns F1 = 19.83, NoAns F1 = 97.76, macro-QA F1 = 58.80, and D1-D6 mapping macro-F1 = 0.816. These results indicate that conservative no-answer rejection transfers better than answerable-clause extraction, but the

TABLE 21. Expert-validated IFRS 16 indicator mapping results.

Indicator	Meaning	Accuracy	Precision	Recall	F1
D1	ROU asset capitalization	0.884	0.891	0.872	0.881
D2	Short-term lease exemption	0.862	0.848	0.835	0.841
D3	Low-value asset exemption	0.846	0.821	0.803	0.812
D4	Renewal option	0.817	0.798	0.781	0.789
D5	Variable lease payments	0.853	0.842	0.816	0.829
D6	Discount-rate disclosure	0.908	0.914	0.887	0.900
Macro average	-	0.862	0.852	0.832	0.842

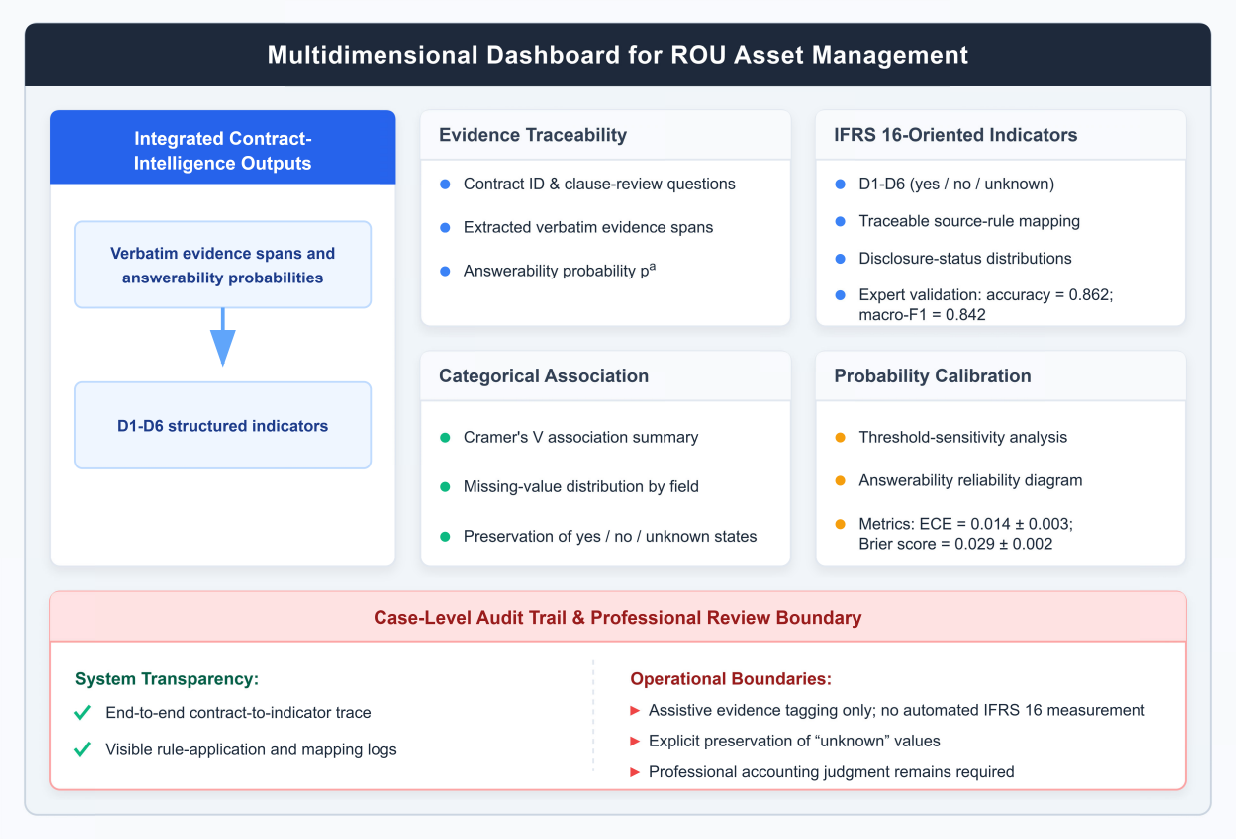


FIGURE 10. Multidimensional dashboard for evidence traceability, categorical association analysis, probability calibration, and IFRS 16-oriented indicator support.

TABLE 22. Expert agreement for IFRS 16 indicator annotation.

Annotation setting	Metric	Value
Two accounting experts	Cohen's kappa	0.78
Three accounting / audit annotators	Fleiss' kappa	0.74
After expert adjudication	Agreement	0.91

external evidence remains limited in scale and should be interpreted descriptively. Because this set is independent of CUAD, the 3.07-point aggregate-F1 decrease is not analyzed with a paired test. Table 24 reports the subset-level and overall external results.

Runtime analysis shows that ROU-TransQA-v2 required 94 minutes for five epochs, 36 ms per QA instance, throughput of 27.8 instances per second, and 9.4 GB GPU memory. Encoder runtime is compared under the same hardware. Autoregressive decoder latency is not mixed with encoder latency because generation length and QLoRA create a different computational regime. Table 25 reports the encoder runtime and inference comparison.

User-interface testing also revealed a context-density shift. CUAD contains long-form legal context, whereas short synthetic snippets are semantically compressed. The neural model is therefore evaluated on long-form contract instances, while short snippets are restricted to rule-validation demonstrations. Figure 10 presents the corrected dashboard terminology, including Cramer's V associations and separate threshold-sensitivity and probability-calibration panels.

N. ERROR ANALYSIS

A stratified manual audit was conducted on 1,000 incorrectly answered answerable test instances. Each instance was assigned to one mutually exclusive primary category by two

TABLE 23. Comparison of IFRS 16 rule-mapping versions.

Rule version	Accuracy	Macro-F1	D1-F1	D2-F1	D3-F1	D4-F1	D5-F1	D6-F1
Rule-v1: keyword-only mapping	0.741	0.704	0.762	0.701	0.682	0.641	0.694	0.744
Rule-v2: enhanced IFRS logic	0.823	0.792	0.841	0.783	0.761	0.734	0.779	0.854
Rule-v3: expert-refined mapping	0.862	0.842	0.881	0.841	0.812	0.789	0.829	0.900

TABLE 24. Descriptive external enterprise-contract validation.

External dataset	Contracts	QA pairs	EM	Overall F1	HasAns F1	NoAns F1	D1-D6 macro-F1	Macro-QA F1
Set A	20	820	71.36	72.08	20.16	97.94	0.824	59.05
Set B	20	820	69.84	70.51	18.92	97.31	0.807	58.12
Set C	20	820	71.28	71.96	20.41	98.02	0.816	59.22
Overall	60	2,460	70.83	71.53	19.83	97.76	0.816	58.80

TABLE 25. Runtime and inference latency comparison.

Model	Training time / epoch	Total training time	Inference latency / QA pair	Throughput	GPU memory
DistilRoBERTa-QA	9.8 min	49 min	18 ms	55.6 samples/s	5.2 GB
RoBERTa-base-QA	16.4 min	82 min	31 ms	32.3 samples/s	8.7 GB
RoBERTa-large-QA	42.7 min	214 min	59 ms	16.9 samples/s	15.8 GB
Legal-BERT-QA	17.1 min	86 min	34 ms	29.4 samples/s	8.9 GB
Longformer-base-QA	38.6 min	193 min	72 ms	13.9 samples/s	14.6 GB
DeBERTa-v3-base-QA	24.3 min	122 min	46 ms	21.7 samples/s	11.8 GB
ROU-TransQA-v2	18.8 min	94 min	36 ms	27.8 samples/s	9.4 GB

reviewers, with disagreements adjudicated. The audit identified 346 missed extractions, 258 boundary overextensions or underextensions, 187 semantically related but incomplete spans, 124 long-context or cross-window evidence losses, and 85 negation, exception, or conditional-language errors. Table 26 reports the mutually exclusive answerable-instance error categories.

TABLE 26. Error distribution on answerable CUAD samples.

Primary error category	Count	Proportion
Missed extraction	346	34.6%
Boundary overextension or underextension	258	25.8%
Semantically related but incomplete span	187	18.7%
Long-context or cross-window evidence loss	124	12.4%
Negation, exception, or conditional-language error	85	8.5%

Among 2,852 unanswerable test instances, 2,834 were correctly rejected and 18 produced false-positive spans. Manual review attributed 12 false positives to lexical-overlap errors and 6 to topically related but non-supporting clauses. This asymmetry confirms that the model is substantially stronger at rejecting absent clauses than at recovering complete evidence for present clauses. Tables 27 and 28 report the unanswerable-instance outcomes and representative error cases, respectively.

TABLE 27. Error distribution on unanswerable CUAD samples.

Unanswerable-instance outcome	Count	Proportion
Correct no-answer	2,834	99.37%
False-positive span: lexical overlap	12	0.42%
False-positive span: related but non-supporting clause	6	0.21%
All false-positive spans	18	0.63%

O. SUMMARY OF RESULTS

First, a span-only extractive model is unsuitable for this high-abstention contract-review setting because it frequently forces unsupported answers. Second, explicit answerability modeling produces the dominant aggregate gain, whereas conditional masking and threshold tuning provide smaller refinements. Third, recent instruction-tuned and legal-domain models recover answerable clauses more effectively, while ROU-TransQA-v2 provides the strongest conservative rejection. Fourth, expert mapping and categorical association analysis support traceable IFRS 16-oriented analytics without converting uncertainty into false certainty.

The independent-test, calibration, ablation, paired-bootstrap, external-validation, and error-analysis results consistently show that overall F1 must not be interpreted as a uniform span-extraction improvement. Aggregate performance is dominated by the unanswerable majority, and macro-QA F1 is therefore reported alongside overall F1.

External validation and runtime analysis indicate preliminary deployment feasibility but also reveal domain shift and persistent answerable-span weakness. The resulting platform is an assistive, traceable, and answerability-aware evidence system rather than a fully automated IFRS 16 measurement engine.

V. DISCUSSION

The results demonstrate that explicit no-answer control is operationally valuable but also narrow the interpretation of the method. The span-only baseline recovered some answerable evidence but frequently produced unsupported spans. ROU-TransQA-v2 reversed this behavior through a separate answerability output and conditional span supervision, yielding near-ceiling no-answer performance. However, answerable-span extraction remained weak, with

TABLE 28. Representative error-analysis cases.

Case type	Question / clause focus	Gold behavior	Predicted behavior	Interpretation
Correct extraction	Governing-law clause	Return the governing-law sentence.	Returned the correct sentence and marked answerable.	Structured legal clauses with clear lexical cues are easier to recover.
False positive	Renewal option absent	Return no-answer.	Returned a semantically related termination or extension sentence.	The model may confuse related commercial provisions with the queried clause.
Missed extraction	Variable payment clause present	Return the variable-rent span.	Returned no-answer.	Long or dispersed payment language reduces answerable-span recall.
Missed answerable evidence	Low-value or exemption wording present	Return the exemption evidence.	Predicted no-answer.	Conditional exemption language and thresholds remain difficult.

independent-test HasAns F1 = 21.64 and five-seed HasAns F1 = 21.54 ± 0.38. Saul-7B-Instruct-v1 achieved higher HasAns F1 (45.92) and macro-QA F1 (67.28), whereas ROU-TransQA-v2 achieved higher NoAns F1 (99.36) and answerability F1 (94.85). The framework should therefore be understood as a conservative and traceable evidence-management workflow rather than as a universally superior QA model.

Threshold selection and probability calibration address different deployment requirements. The validation threshold implements a risk-sensitive answer-return policy, while temperature scaling improves the correspondence between answerability probabilities and empirical frequencies. Operational thresholds may be adjusted according to the relative cost of unsupported evidence and abstention, but any deployment-specific change should be validated on held-out contracts.

The system-level contribution lies in linking source evidence, answerability risk, deterministic mapping rules, yes/no/unknown indicator states, and audit-oriented visualization. These functions remain useful even when a larger instruction-tuned model provides stronger span recovery because every indicator must remain traceable to a contract clause and an explicit rule. Recent industry-specific evidence confirms that IFRS 16 effects vary with reporting context and analytical scope [40].

The large number of unknown disclosure values confirms that uncertainty should be preserved. Unknown denotes insufficient evidence rather than a negative disclosure and should trigger review rather than automatic imputation. This distinction is maintained in the data model, rule layer, visualizations, and expert-validation protocol.

Several limitations define the scope of the contribution. First, the neural component is a task-specific adaptation and integration of established QA mechanisms rather than a fundamentally new Transformer architecture. Second, answerable-span extraction remains substantially weaker than no-answer rejection, and recent instruction-tuned legal models achieve higher HasAns F1 and macro-QA F1. Third, decoder comparisons are sensitive to prompt construction, quantization, window aggregation, output normalization, and the contiguous-source-span constraint. Fourth, the D1-D6 mapping layer is expert-informed and may require entity- and jurisdiction-specific adaptation. Fifth, the evidence is derived mainly from English CUAD-style contracts, a modest IFRS 16 disclosure dataset, and three anonymized enterprise subsets containing 60 contracts and 2,460 QA instances; the external HasAns F1 of 19.83 indicates that answerable-clause extraction remains fragile under domain shift. Finally, the system does not calculate lease liabilities, value right-of-use assets, prepare amortization schedules, or replace professional accounting judgment. These boundaries are consistent with the broader IFRS-adoption literature, which emphasizes institutional and jurisdictional heterogeneity [41].

Future work should combine retrieval-enhanced long-context extraction, calibrated selective prediction, multilingual and cross-jurisdiction evaluation, expert-refined rule libraries, and human-in-the-loop enterprise validation. Larger legal foundation models should be evaluated under the same source-evidence and abstention constraints rather than only through unconstrained generation.

VI. CONCLUSION

This study developed an answerability-aware contract-intelligence workflow for IFRS 16 right-of-use asset management. The trainable ROU-TransQA-v2 component separates clause-presence prediction from span localization and excludes unanswerable instances from direct span-loss supervision. Across five contract-level runs, it achieved independent-test EM = 73.92 ± 0.30, overall F1 = 74.60 ± 0.30, HasAns F1 = 21.54 ± 0.38, NoAns F1 = 99.34 ± 0.12, and macro-QA F1 = 60.44 ± 0.20. Because unanswerable instances account for 68.20% of the independent test set, this aggregate performance should be interpreted primarily as conservative no-answer control rather than uniformly strong span extraction. Comparisons with recent legal-domain and instruction-tuned models show that Saul-7B-Instruct-v1 achieved stronger HasAns F1 and macro-QA F1, whereas ROU-TransQA-v2 retained stronger no-answer rejection and deterministic evidence traceability. External validation over 60 enterprise contracts and 2,460 QA instances yielded overall F1 = 71.53, HasAns F1 = 19.83, NoAns F1 = 97.76, and D1-D6 mapping macro-F1 = 0.816, indicating practical transfer of abstention behavior but persistent weakness in answerable evidence recovery. Expert validation of the D1-D6 mapping layer yielded accuracy = 0.862 and macro-F1 = 0.842. The framework consequently provides an

assistive bridge from contract evidence to accounting-oriented analytics while preserving the need for professional review and the distinction between evidence tagging and IFRS 16 measurement.

DATA AVAILABILITY

The CUAD contract-QA benchmark, which is not an IFRS 16-specific dataset, is publicly available from the official CUAD release (CUAD dataset link). The structured IFRS 16 disclosure dataset described by Tumpach et al. is available through Mendeley Data (IFRS 16 disclosure dataset link). Raw enterprise contracts used for external validation cannot be publicly distributed because of contractual confidentiality; de-identified aggregate results and derived annotation records are shared. The fixed contract-level split manifest, processed yes/no/unknown indicator labels, prompt templates, and de-identified prediction summaries are included in the anonymized reproducibility package.

CODE AVAILABILITY

The source code and associated reproducibility materials for ROU-TransQA-v2 are available in the anonymous repository at: (https://anonymous.4open.science/status/ROU-TransQA-v2-Anonymous-Reproducibility-1515) The repository contains the implementation, configuration files, evaluation and analysis scripts, unit tests, environment specifications, and documentation for the computational workflow.

AUTHOR CONTRIBUTIONS STATEMENT

Conceptualization, Yingying Han and Li Huang; Methodology, Yingying Han and Ye Wang; Investigation, Yingying Han and Ye Wang; Writing–original draft preparation, Yingying Han; Writing–review and editing, Yingying Han, Ye Wang, and Li Huang; Supervision, Li Huang. All authors have read and agreed to the published version of the manuscript.

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